

MetaITEM NUMBER: CM-01979

REV:	DESCRIPTION:	ECO:	DATE:	APPROVED:
0.1	SENT TO CUSTOMER FOR REVIEW		09/09/2024	J. PARKER
0.2	DESIGN REWORK SENT TO CUSTOMER FOR REVIEW		12/13/2024	J. PARKER
0.3	REGISTER AND COMMAND UPDATES		06/01/2025	J. PARKER
0.4	BOOT MODE, FDIR DETAILS, REGISTER UPDATES AND COMMAND UPDATES		06/17/2025	J. PARKER
0.5	MINOR REGISTER UPDATES INCLUDING ADDITION OF IP and MAC REGISTERS FOR CONFIGURING THE SECOND ETHERNET AS WELL AS A PROASIC3 FIRMWARE VERSION REGISTER		7/15/2025	J. PARKER
0.6	SIMPLIFIED TMTC AND REMOVAL OF EVENTS FOR INITIAL RELEASE			

ITAR EXPORT CONTROLLED - This document contains technical data that is controlled under the U.S. International Traffic in Arms Regulations ("ITAR"), 22 C.F.R. Parts 120-130, and may not be exported, reexported or transferred to a non-U.S. Person except as licensed or otherwise authorized by the U.S. Government. Violations of the ITAR may be subject to both criminal and administrative penalties under the Arms Export Control Act of 1976 (as amended).

TABLE OF CONTENTS

1.0 INTRODUCTION	3
1.1 PURPOSE.....	4
1.2 SCOPE.....	4
1.3 USAGE.....	4
1.4 CONTROL AUTHORITY	4
2.0 APPLICABLE DOCUMENTS	4
2.1 GOVERNMENT DOCUMENTATION	4
2.2 INTERNAL DOCUMENTATION	4
3.0 FARSIGHT SYSTEM OVERVIEW	5
3.1 TURION MISSION AND COMPANY AND COMPANY PROFILE	5
3.2 FARSIGHT COMPLIANCE.....	5
4.0 OPERATION & ARCHITECTURE	5
4.1 SYSTEM MODES	7
4.1.1.....BOOT MODE (BOOT_MODE)	10
4.1.2..... Low-Power Mode (LP_MODE – 0x00)	11
4.1.3.....Update Mode (UPDATE_MODE - 0x01)	12
4.1.4..... Idle Mode (IDLE_MODE – 0x02)	13
4.1.5..... Manual Imaging Mode (MANUAL_MODE – 0x03)	14
4.1.6..... Streaming Mode (STREAM_MODE – 0x04)	16
4.2 IMAGING OPERATIONS	17
4.2.1.....SETUP AND CONFIGURATION	17
4.2.2..... TRIGGERING OF IMAGE ACQUISITION	18
4.2.3..... TIME SYNCHRONIZATION	18
4.2.4..... Resets & Watchdogs	19

4.2.5.....	Latchup & Boot Failure Response	19
5.0 SOFTWARE TMTC INTERFACE		20
5.1 SERIAL INTERFACE		21
5.2 TMTC PACKET FORMAT		21
5.2.1.....	PKT_HDR	22
5.2.2.....	PKT_DATA	23
5.2.3.....	TL DATAGRAM FORMAT	23
5.3 FARSIGHT COMMANDS		24
5.3.1Error Enumerations		25
5.3.2Command Descriptions		26
5.3.2.1 GET_VERSION_SYS_CMD		27
5.3.2.2 SET_MODE_SYS_CMD		27
5.3.2.3 RESET_SYS_CMD		28
5.3.2.4 REG_READ_SYS_CMD		28
5.3.2.5 REG_WRITE_SYS_CMD		29
5.3.2.6 SET_TIME_SYS_CMD		29
5.3.2.7 START_ACQ_SYS_CMD		30
5.3.2.8 STOP_ACQ_SYS_CMD		30
5.3.2.9 FOCUS_SYS_CMD		30
5.3.2.10 HALT_FOCUS_SYS_CMD [Included in future releases]		31
5.3.2.11 XFER_BUFF_SYS_CMD		31
5.3.2.12 RESET_BUFF_SYS_CMD [Included in future releases]		32
5.3.2.13 GET_INFO_SYS_CMD [Included in future releases]		32
5.3.2.14 CONF_EVENT_LOG_SYS_CMD [Included in future releases]		33
5.3.2.15 EN_TL_CHAN_LOG_SYS_CMD [Included in future releases]		33
5.3.2.16 XFER_LOG_SYS_CMD [Included in future releases]		34
5.4 FARSIGHT REGISTERS		34
5.4.1CAMERA STATUS AND CONTROL REGISTERS		35

5.4.2	SYSTEM REGISTERS.....	35
5.4.3	SYSTEM IMAGING REGISTERS	35
5.4.4	 SYSTEM COMMUNICATION REGISTERS	
	36	
5.4.5	 SYSTEM VERSION REGISTERS	
	36	
5.5	EXTENDED SYSTEM REGISTERS	36
5.6	REGISTER DESCRIPTIONS	37
5.6.1	 SENSOR_STAT_SYS_REG	
	37	
5.6.2	SENSOR_FRM_MOD_SYS_REG	
	37	
5.6.3	SENSOR_BIT_DEPTH_SYS_REG	
	38	
5.6.4	 SENSOR_GAIN_SYS_REG	
	38	
5.6.5	SENSOR_BLO_SYS_REG	
	38	
5.6.6	SENSOR_EXPO_USEC_SYS_REG	
	38	
5.6.7	SENSOR_TEMP_KEL_SYS_REG	
	39	
5.6.8	SENSOR_OVER_TEMP_KEL_SYS_REG	
	39	
5.6.9	MODE_SYS_REG	
	39	
5.6.10	LAST_ERROR_SYS_REG.....	40
5.6.11	RST_REASON_SYS_REG	40
5.6.12	FPGA_TEMP_KEL_SYS_REG	40
5.6.13	FPGA_VOLTAGE_SYS_REG	40
5.6.14	DDR4_16_PWR_STATUS.....	41
5.6.15	DDR4_8_PWR_STATUS.....	41
5.6.16	LVDS_PWR_STATUS.....	41
5.6.17	ETH_1_PWR_STATUS	41
5.6.18	ETH_2_PWR_STATUS	41
5.6.19	FOCUS_MECH_PWR_STATUS.....	41

5.6.20	SENSOR_PWR_STATUS.....	41
5.6.21	ERROR_COUNT_SYS_REG.....	41
5.6.22	BOOT_COUNT_SYS_REG.....	41
5.6.23	UPTIME_MSEC_SYS_REG	42
5.6.24	TIMESTAMP_UNIX_SYS_REG.....	42
5.6.25	WDOG_TIMEOUT_MSEC_SYS_REG	42
5.6.26	BUFF_WRITE_POLICY_SYS_REG	42
5.6.27	BUFF_WRITE_PTR_SYS_REG.....	43
5.6.28	BUFF_BYTES_AVAIL_SYS_REG	43
5.6.29	FRAME_BYTE_PACKING_EN_REG	43
5.6.30	TRIG_MODE_SYS_REG	44
5.6.31	IMG_PER_TRIGGER_SYS_REG.....	45
5.6.32	FRAMES_PER_SEC_SYS_REG	46
5.6.33	CAP_START_DLY_USEC_SYS_REG	46
5.6.34	FRAME_COUNT_SYS_REG.....	46
5.6.35	FOCUS_REQ_DIST_M_SYS_REG.....	46
5.6.36	FOCUS_COMP_DIST_M_SYS_REG	47
5.6.37	FOCUS_LVDT_POS_NM_REG.....	47
5.6.38	GPO_CONF_SYS_REG	47
5.6.39	GPI_CONF_SYS_REG	47
5.6.40	INTR_PIN_CONF_SYS_REG.....	48
5.6.41	TRIG_PIN_CONF_SYS_REG.....	48
5.6.42	TRIG_DELAY_USEC_SYS_REG.....	49
5.6.43	TRIG_TIMEOUT_MSEC_SYS_REG	49
5.6.44	TMTC_TIMEOUT_MSEC_SYS_REG	49
5.6.45	SER_BAUD_SYS_REG.....	49
5.6.46	MAC_ADDRESS_1_SYS_REG	50
5.6.47	MAC_ADDRESS_2_SYS_REG	50
5.6.48	IP_ADDRESS_1_SYS_REG.....	50
5.6.49	IP_ADDRESS_2_SYS_REG.....	50
5.6.50	GATEWAY_SYS_REG	50
5.6.51	SUBNET_MASK_SYS_REG.....	50
5.6.52	ETH_0_PORT_SYS_REG.....	50

5.6.53	ETH_1_PORT_SYS_REG	51
5.6.54	FAV_DEVICE_ID_SYS_REG	51
5.6.55	SENSOR_ID_SYS_REG	51
5.6.56	TMTC_VERS_SYS_REG	51
5.6.57	SW_VERS_SYS_REG	51
5.6.58	PF_FW_VERS_SYS_REG	51
5.6.59	P3_FW_VERS_SYS_REG	52
5.6.60	FW_MD5_SYS_REG.....	52
5.0.....	PCIe DATA TRANSFER	52
6.0.....	IMAGE FORMAT & METADATA	53
6.1	IMAGE FRAME FORMAT	53
6.2	IMAGE METADATA.....	53

LIST OF FIGURES

FIGURE 4-1:	FARSIGHT CAMERA OPERATIONS	11
FIGURE 4-2:	FARSIGHT ARCHITECTURE DIAGRAM.....	12
FIGURE 4-3:	MODE TRANSITION DIAGRAM.....	13
FIGURE 4-4:	LP_MODE - FINITE STATE MACHINE (TBR)	17
FIGURE 4-:	UPDATE_MODE-FINITE STATE MACHINE (TBR).....	18
FIGURE 4-:	IDLE_MODE – FINITE STATE MACHINE (TBR)	19
FIGURE 4-:	MANUAL_MODE – FINITE STATE MACHINE (TBR)	20
FIGURE 4-:	STREAM_MODE – FINITE STATE MACHINE (TBR)	21
FIGURE 5-1:	TMTC PACKET FORMAT.....	26
FIGURE 5-2:	TL EVENT, TL CHANNEL & RAW IMAGE SEGMENT DATAGRAM FORMATTING	28

LIST OF TABLES

TABLE 2-1:	APPLICABLE GOVERNMENT DOCUMENTATION	8
TABLE 2-2:	APPLICABLE TURION DOCUMENTATION	9
TABLE 4-1:	MODES OF OPERATION	12
TABLE 4-2	BOOT_MODE-ENTRANCE & EXIT CRITERIA (TBR)	15
TABLE 4-3:	LP_MODE-ENTRANCE & EXIT CRITERIA (TBR)	16
TABLE 4-4:	UPDATE_MODE - ENTRANCE & EXIT CRITERIA (TBR)	17
TABLE 4-5:	IDLE_MODE-ENTRANCE & EXIT CRITERIA (TBR)	18
TABLE 4-6:	MANUAL_MODE – ENTRANCE & EXIT CRITERIA (TBR)	19

TABLE 4-7: STREAM_MODE – ENTRANCE & EXIT CRITERIA (TBR)	20
TABLE 5-1: SOFTWARE TMTC INTERFACE TERMINOLOGIES	24
TABLE 5-2: FARSIGHT’S SERIAL DEBUG AND BACKUP TMTC INTERFACE QUICK REFERENCE	24
TABLE 5-3: PACKET HEADER FIELD DESCRIPTIONS	26
TABLE 5-4: TELEMETRY (TL) EVENT DESCRIPTIONS	28

LIST OF ACRONYMS/ABBREVIATIONS

ACRONYM/ABBREVIATION	DEFINITION
C&DH	Command and Data Handling
CDS	Correlated Double Sampling
CMOS	Complementary Metal-Oxide-Semiconductor
CONOPS	Concept of Operations
CSR	Control and Status Registers
DEDOO	Dedicated Input/Output
EC	Embedded Clock
ECO	Engineering Change Order
FAV	FARSIGHT Payload Avionic Computer
FC	Flight Computer
FD	Full Digital
FPGA	Field Programmable Gate Array
GEO	Geostationary Orbit
GMRWT	Global Master Read Wait Time
GMTWT	Global Master Transfer Wait Time
GPI	General Purpose Input
GPO	General Purpose Output
HAL	FAV Hardware Abstraction Layer Middleware
ICD	Interface Control Document
LEO	Low Earth Orbit
NVM	Non-Volatile Memory
OB	Optical Black
PC	Payload Computer
PCIESS	Embedded PCIe Controller Subsystem
PGA	Programmable Gain Amplifier
PPS	Pulse Per Second
PWR	Power
RoHS	Restriction of Hazardous Substances

SERDES	Serializer/De-serializer
SHS	Shutter Sweep Time
SLVS	Scalable Low-Voltage Signaling
SLVS-EC	Scalable Low-Voltage Signaling with Embedded Clock
TMTC	Telemetry and Telecommand

1.0 INTRODUCTION

This document defines the interfaces and configuration parameters of the FARSIGHT system firmware. ~~as a standalone camera module.~~

1.1 PURPOSE

This document serves as an overview of the software interface for the FARSIGHT telescope. This Interface Control Document (ICD) is not a standalone document and must be used in conjunction with the documents listed in Section 2 to ensure successful payload operations.

1.2 SCOPE

This document is valid with respect to the FARSIGHT system as a standalone camera system.

1.3 USAGE

This document is to be used by users of the FARSIGHT payload for development of software to interface with and operate the FARSIGHT payload as a standalone camera.

1.4 CONTROL AUTHORITY

Upon release of version 1.0, this document is under the TURION quality control system and shall propose changes in accordance with the internal Engineering Change Order (ECO) process. Prior to version 1.0 this document is considered under a DRAFT status.

2.0 APPLICABLE DOCUMENTS

The following documents, both program and non-program, are identified within this document. Unless a version number is specified, the latest version of these documents is to be used.

2.1 GOVERNMENT DOCUMENTATION

Future revisions of FARSIGHT's Software ICD will include modifications to ensure compliance with the Government Standards listing in Table 2.1: Applicable Government Documentation.

Table 2-1: Applicable Government Documentation

Identifier	Document Title
-	JCO, SSA, GUIDANCE, CORRECTIONS_JULY2019.pdf
-	NGA.STND.0067_1.0_MIDS_2021-10-20.pdf
-	NATIONAL IMAGERY TRANSMISSION FORMAT 2.1: Motion Imagery Extension, version 1.3.2, 2021-10-20

2.2 INTERNAL DOCUMENTATION

The following TURION documents and authorized changes form a part of this document.

Table 2-2: Applicable TURION Documentation

Identifier	Document Title
CM-00945	EXPORT CONTROL, GUIDELINES, DOCUMENT MARKING
CM-01980	ICD, ELECTRICAL, FARSIGHT

2.3 SUPPLIER DOCUMENTATION

The following external supplier documents are referenced throughout the document.

SUPPLIER	Document Title
Microchip	Polarfire MPF500T Specification, 2022. PolarFire-FPGA-Datasheet-DS00003831
Microchip	VSC8541-02/VSC8451-05 Datasheet, 2023
Microchip	Coretse Handbook, 2024.
Microchip	DDR AXI4 Arbiter User Guide, 2024.
Microchip	PolarFire FPGA SLVS-EC User Guide, 2024.
Microchip	SPI-DIRECTC User Guide, 2022.
Sony	IMX531-AAMJ-C Camera Specification, 2022. E20521D2Z.

3.0 FARSIGHT SYSTEM OVERVIEW

FARSIGHT is an optical payload designed for mid- to long-range non-Earth Imaging (NEI), supporting 5-year missions in Low-Earth Orbit (LEO) and Geostationary Orbit (GEO). Within the context of this document and based on the current software interface design, FARSIGHT represents a standalone camera.

3.1 TURION MISSION AND COMPANY AND COMPANY PROFILE

TURION was founded to accelerate the commercialization of space while focusing on key technological risks toward advancement of the greater vision of a prosperous and stable space economy. To address these risks, TURION uses a hardware-first approach to rapidly develop and iterate on space vehicles using a capability-based architecture to reduce risk and accelerate design iteration timelines.

3.2 FARSIGHT COMPLIANCE

FARSIGHT must enable the completion of the DRMs set forth by the relevant spacecraft mission. The SW version flown on each spacecraft must be compliant with restrictions or capabilities set forth by the DRMs.

3.3 FARSIGHT CAMERA SENSOR

FARSIGHT uses a Sony IMX531 camera sensor for capturing non-earth images. The specification of the camera sensor is provided in this section.

3.3.1 IMX531 Camera Sensor Features

The IMX531-AAMJ-C is a diagonal 17.5 mm (Type 1.1) CMOS active pixel type solid-state image sensor with a square pixel array and 20.28 M effective pixels. This chip features a global shutter with variable charge-integration time.

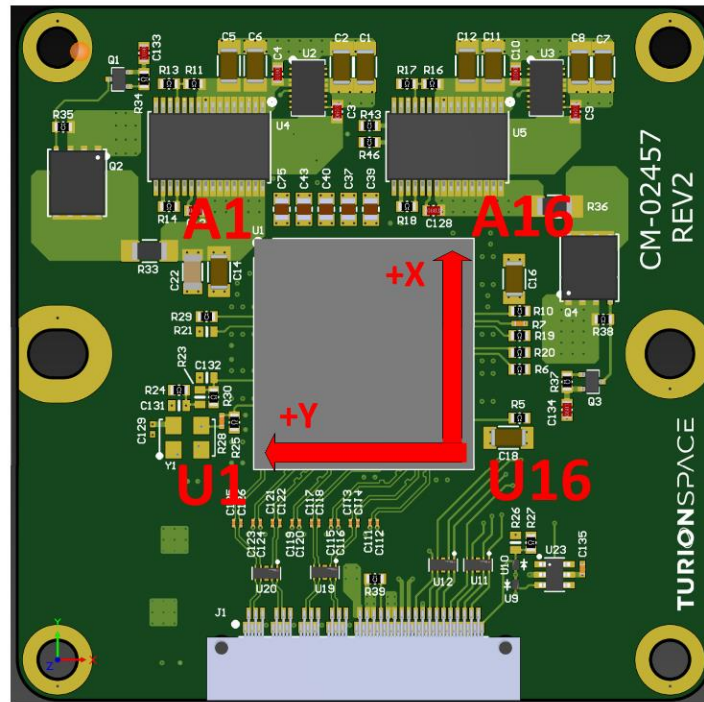
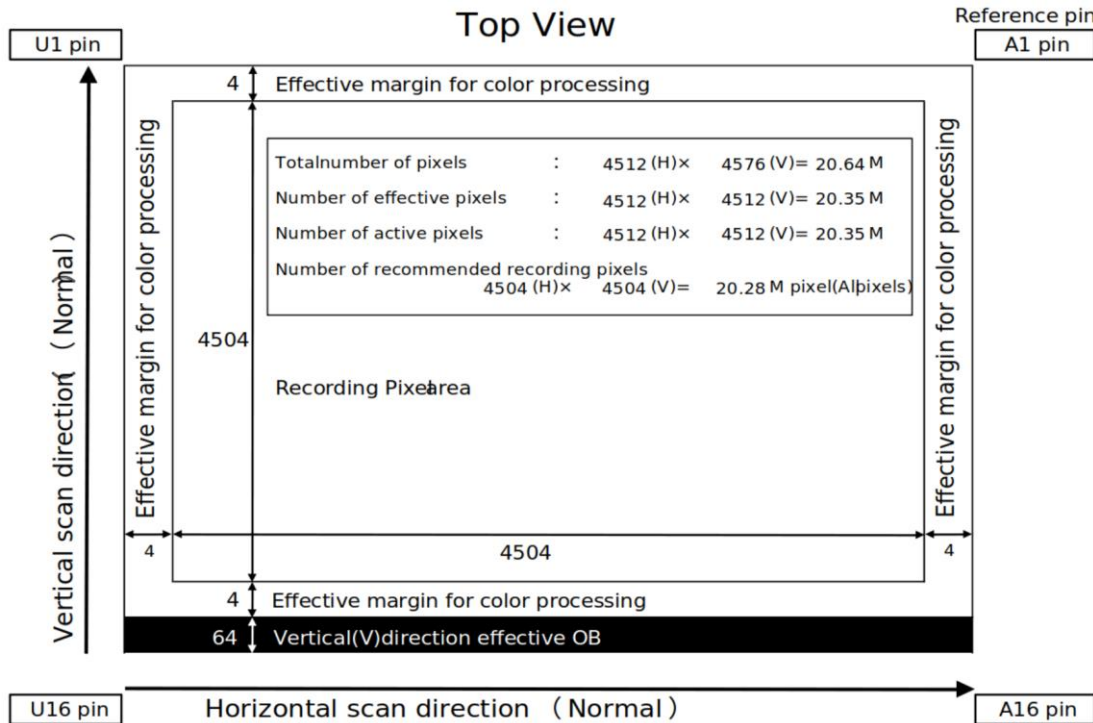
- CMOS active pixel type dots
- Built-in timing adjustment circuit
- Horizontal and vertical driver
- Serial communication circuit
- Global shutter function
- Embedded clock frequency: 54 MHz
- Number of recommended recording pixels: 4504(H) X 4504(V), approx. 20.28 MP

Features supported in FARSIGHT Avionics Firmware:

- All-pixel scan mode
- 12-bit pixel width
- Max frame rate 64 fps for 8GB DDR4 and 70 fps for 16 GB DDR4 memory buffer
- Variable-exposure duration via controlled trigger pin
- Monitor output for exposure period via dedicated output pin
- I/O interface SLVSEC 8 lane output
- Data transfer rate at 4.752 Gbps
- Programmable Gain Amplifier:
 - 0 dB to 24 dB: Analog Gain (0.1 dB step)
 - 24.1 dB to 48 dB: 24 dB Analog Gain + Digital Gain: 0.1 dB to 24 dB (0.1 dB step)

3.3.2 IMX531 Camera Sensor Pixel Arrangement

The recommended recording pixels of the camera sensor is $4504 \times 4504 = 20.28$ MP. The details of the pixel array are shown in **FIGURES REF.**



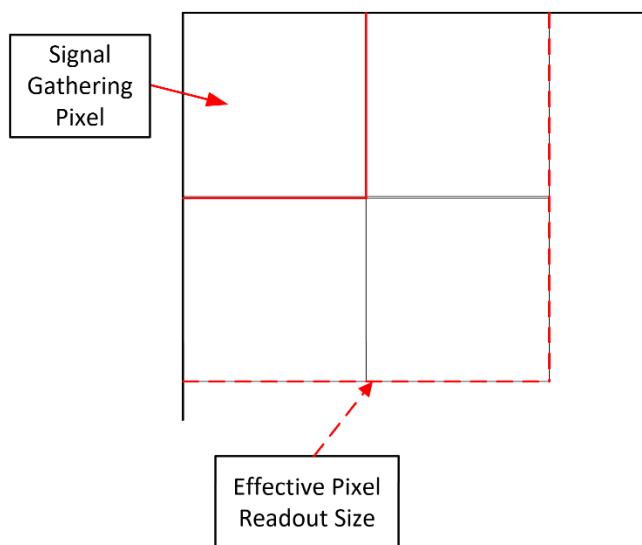
3.3.3 IMX531 Camera Sensor Readout Modes

The IMX531 image sensor supports four different readout modes. The readout modes are:

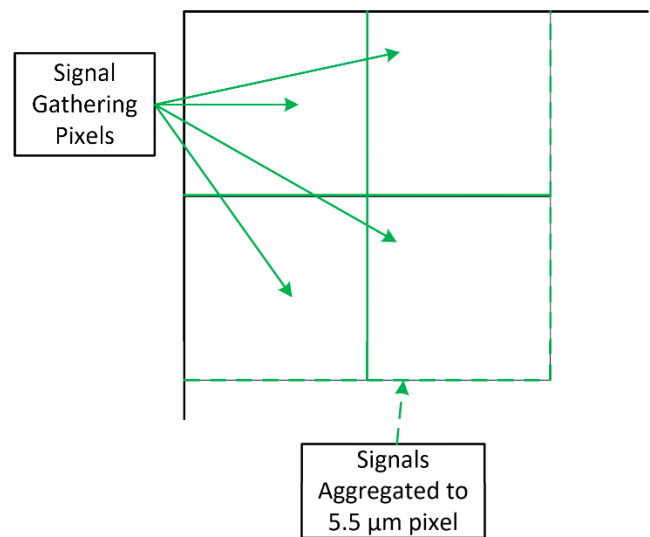
1. All Pixel Scan

- a. All pixels on the sensor are read without modification
2. Vertical/Horizontal $\frac{1}{2}$ subsampling
 - a. Pixels are sampled at $\frac{1}{2}$ rate. Output image is 2252 x 2252.
3. 2x2 FD binning
 - a. 2x2 grids of pixels and their respective photodiode are aggregated into one pixel. Output image is 2252 x 2252
4. Region of Interest (ROI) and Overlapped ROI
 - a. Selective regions of the sensor can be chosen to be read. Image readout size depends on commanded ROI.

Vertical/Horizontal $\frac{1}{2}$ Sampling



2x2 FD Binning

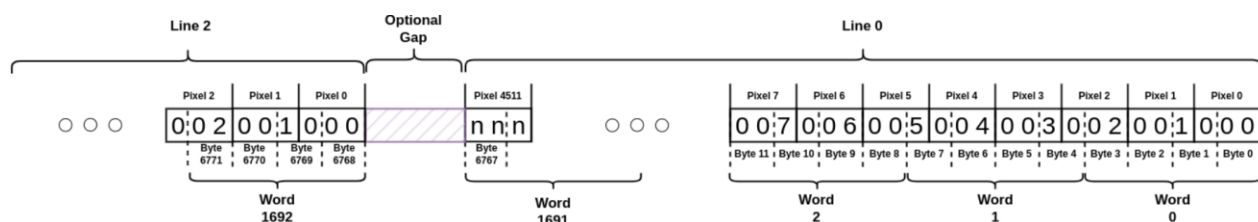


Readout modes are only valid during operation modes. Meaning there won't be any reading during standby mode.

In the current version of FARSIGHT only the All-Pixel scan mode is supported.

3.3.4 FARSIGHT PACKED PIXEL STORAGE FORMAT

Figure XXX illustrates how pixels are stored in the DDR4 memory space. Pixels are stored in ascending order, starting at the lowest memory address of the DDR4. In the current design, pixels are always 12 bits wide and stored in a packed format; therefore, unpacking to 16 bits is required before further processing of the image.



The first line of the frame always contains the metadata generated by the FPGA system at the moment of frame capture. Details of the metadata format are discussed in the subsequent sections. The next 64 lines of the frame constitute the Vertical Effective Optical Black (OB) region, which is used for calibration and dark current compensation. The actual image data begins at line 65 and spans a total of 4512 lines. After the 64 lines of VOPB there is a 4 line of effective margin for color processing at the beginning of the image and there is another 4 lines of effective margin for color processing at the end of the image. Each line of the frame consists of 4512 pixels, and since each pixel is 12 bits wide, the total size per line is $4512 \times 12/8 = 6768$ bytes.

3.3.5 FARSIGHT PACKED PIXEL UDP OUTPUT FORMAT

The FPGA transmits image frames via UDP packets to a host computer. Each frame consists of 4512×4576 pixels, where the image data begins at line 65 (after the 64-line Optical Black region and 1 metadata line).

FARSIGHT UDP Packet Structure:

Each UDP packet contains a 6-byte header followed by the pixel payload. Table **XXX** describes the packet header format.

Field	Size (bytes)	Description
Start Marker	2	Packet delimiter
Line Sequence	2	Line number within the frame (big-endian)
Packet Sequence	2	Packet number within the line (big-endian)
Payload	Variable	Packed 12-bit pixel data

Each line is transmitted across 5 UDP packets (packets 0-4), resulting in a total of $4576 \times 5 = 22,880$ packets per frame.

Camera Sensor Row Format:

Each row contains 4512 pixels in packed 12-bit format, yielding $4512 \times 12 / 8 = 6768$ bytes of pixel data. Additionally, each row includes 16 bytes of padding for memory alignment purposes, resulting in a total row size of $6768 + 16 = 6804$ bytes.

The expected total frame size can be calculated as:

$$\text{Frame Size} = \text{Height} \times (\text{Packed Row Bytes} + \text{Padding}) = 4576 \times (6768 + 16) = 31,039,484 \text{ bytes}$$

Camera Sensor 12-bit Packed Pixel Unpacking Instruction

Pixels are stored in a packed format where every 12 bytes (96 bits) encode 8 pixels. The unpacking process interprets the data as three 32-bit big-endian words (W0, W1, W2) and extracts the 8 pixels using bitwise operations as shown in Table 47.

To convert 12-bit pixel values to 16-bit format for standard image processing applications, the pixel values are shifted left by 4 bits:

$$P_{16} = P_{12} \ll 4$$

Image Data	Checksum placeholder	Start Marker	Line Sequence	UDP Packet Sequence
00 24 9b 8b ee fb 00 04	a3 12	34 56	08 00	45 00
05 ce 00 01 00 00 40 11	40 d6	0a 65	0f c0	0a 65
0f bf 04 d2 89 31 05 ba	00 00	50 4b	00 10	00 02
30 00 04 50 00 06 70 00	08 90	00 0a	b0 00	0c d0
00 0e f0 00 10 10 01 12	30 01	14 50	01 16	70 01
18 90 01 1a b0 01 1c d0	01 1e	f0 01	20 10	02 22
30 02 24 50 02 26 70 02	28 90	02 2a	b0 02	2c d0
02 2e f0 02 30 10 03 32	30 03	34 50	03 36	70 03
38 90 03 3a b0 03 3c d0	03 3e	f0 03	40 10	04 42
30 04 44 50 04 46 70 04	48 90	04 4a	b0 04	4c d0
04 4e f0 04 50 10 05 52	30 05	54 50	05 56	70 05
58 90 05 5a b0 05 5c d0	05 5e	f0 05	60 10	06 62

Pixel	Extraction Formula
P_0	$W_0 \wedge 0xFFF$
P_1	$(W_0 \gg 12) \wedge 0xFFF$
P_2	$((W_0 \gg 24) \vee (W_1 \ll 8)) \wedge 0xFFF$
P_3	$(W_1 \gg 4) \wedge 0xFFF$
P_4	$(W_1 \gg 16) \wedge 0xFFF$
P_5	$((W_1 \gg 28) \vee (W_2 \ll 4)) \wedge 0xFFF$
P_6	$(W_2 \gg 8) \wedge 0xFFF$
P_7	$(W_2 \gg 20) \wedge 0xFFF$

Payload Computer Receiver Configuration Requirements:

To reliably receive high-bandwidth UDP image data without packet loss, the host system must be configured with adequate UDP receive buffer sizes. The recommended minimum buffer size is 256 MB. On Linux systems, this can be configured using:

```
sudo sysctl -w net.core.rmem_max=268435456
```

```
sudo sysctl -w net.core.rmem_default=268435456
```

To make these changes persistent across reboots, add the following lines to `/etc/sysctl.conf`:

```
net.core.rmem_max = 268435456
```

```
net.core.rmem_default = 268435456
```

FARSIGHT UDP Packet Loss Handling:

Due to the nature of UDP transmission, packets may occasionally be lost or arrive out of order. The receiver software implements the following strategies to handle these scenarios:

- **Packet Reordering:** Packets are buffered and written in the correct sequence based on line and packet sequence numbers.
- **Timeout Mechanism:** If a packet does not arrive within a configurable wait time (default: 10 ms), the missing data is filled with zeros.
- **Frame Timeout:** If a complete frame is not received within the frame timeout period (default: 2 seconds), the frame is finalized with any missing packets filled with zeros

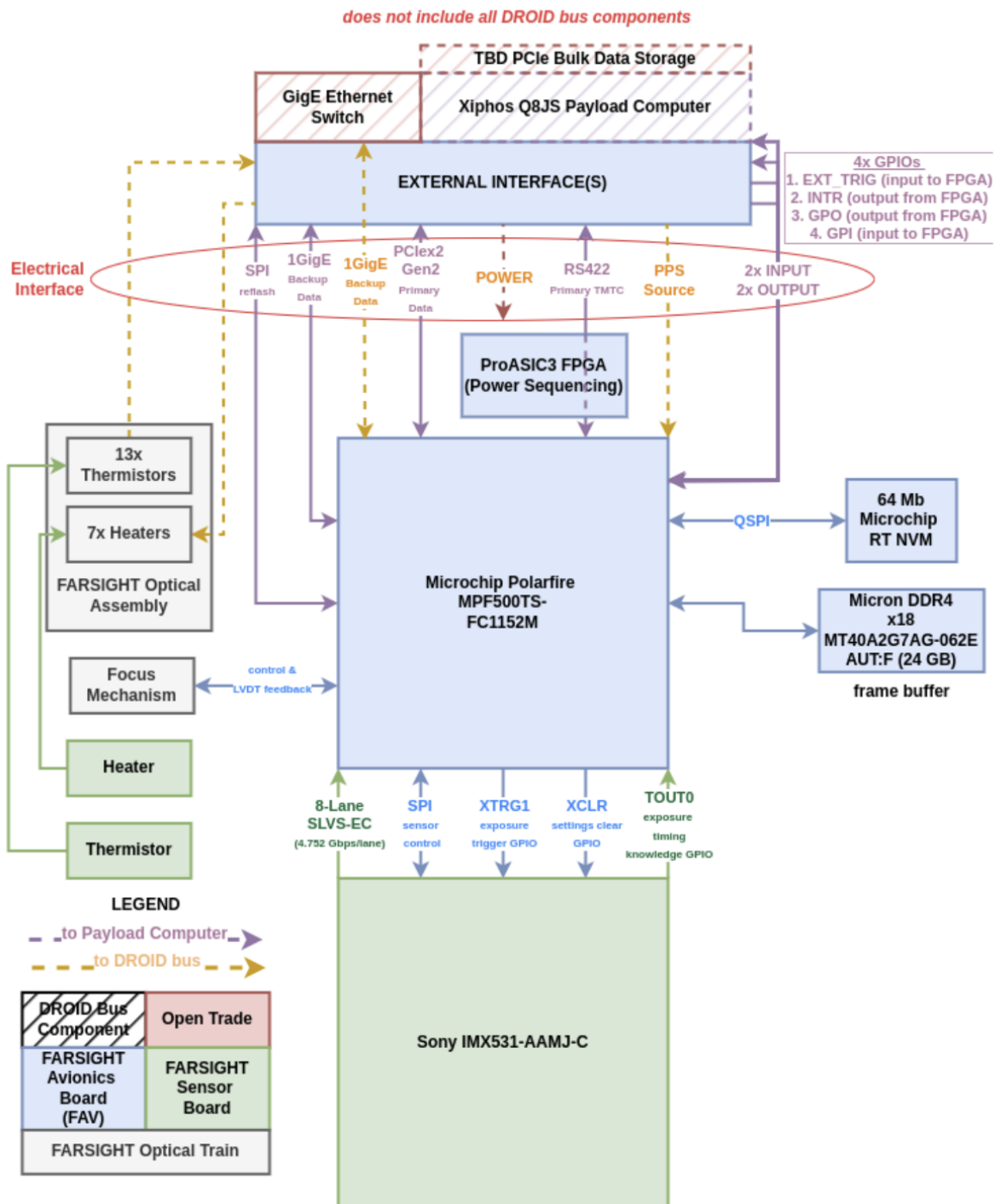
3.4 FARSIGHT HARDWARE ARCHITECTURE

3.4.1 FARSIGHT PHYSICAL INTERFACES

Figure Ref shows the physical interfaces of the FARSIGHT Payload system.

The FARSIGHT system connects to or includes the following components:

- **Xiphos Q8JS Payload Computer (External Components):**
 - **RS422:** Primary TMTC (Telemetry and Telecommand) serial interface.
 - **1GigE (x2):** Dual Gigabit Ethernet connections for backup and primary data transfer.
 - **PCIe x2 Gen2:** High-speed PCIe interface for primary data transfer.
 - **SPI:** Used for reflash operations.
 - **PPS:** Pulse Per Second timing signal source.
 - **GPIOs (4x):** External trigger input (EXT_TRIG), interrupt output (INTR), and general-purpose I/O signals.
- **ProASIC3 FPGA:** Handles power sequencing and system monitoring
- **PolarFire FPGA (MPF500TS-FC1152M):** The main processing unit that interfaces with the NVM and DDR4
 - **64 Mb Microchip RT NVM:** Non-volatile memory connected via QSPI for configuration storage.
 - **Micron DDR4 (24 GB):** Frame buffer memory (18x MT40A2G7AG-062E AUT:F) for image storage.
- **Sony IMX531-AAMJ-C Sensor:** Connected to the PolarFire FPGA
 - **8-Lane SLVS-EC:** High-speed data interface running at 4.752 Gbps/lane
 - **SPI:** Sensor control interface
 - **XTRG1:** Exposure trigger GPIO
 - **XCLR:** Settings clear GPIO
 - **TOUT0:** Exposure timing knowledge GPIO.
- **FARSIGHT Optical Assembly:** Includes 13x thermistors for temperature monitoring and 7x heaters for thermal control.
- **Focus Mechanism:** Controlled by the PolarFire FPGA with LVDT feedback for position sensing.
- **Heater and Thermistor:** Additional thermal management components with direct connection to electrical interface.



4.0 OPERATION & ARCHITECTURE

Figure 4-1 is a high-level flow diagram that provides an example of how FARSIGHT is expected to be operated as a standalone camera module. The rest of this document details modes, commands and telemetry values used to enable this outlined operation.

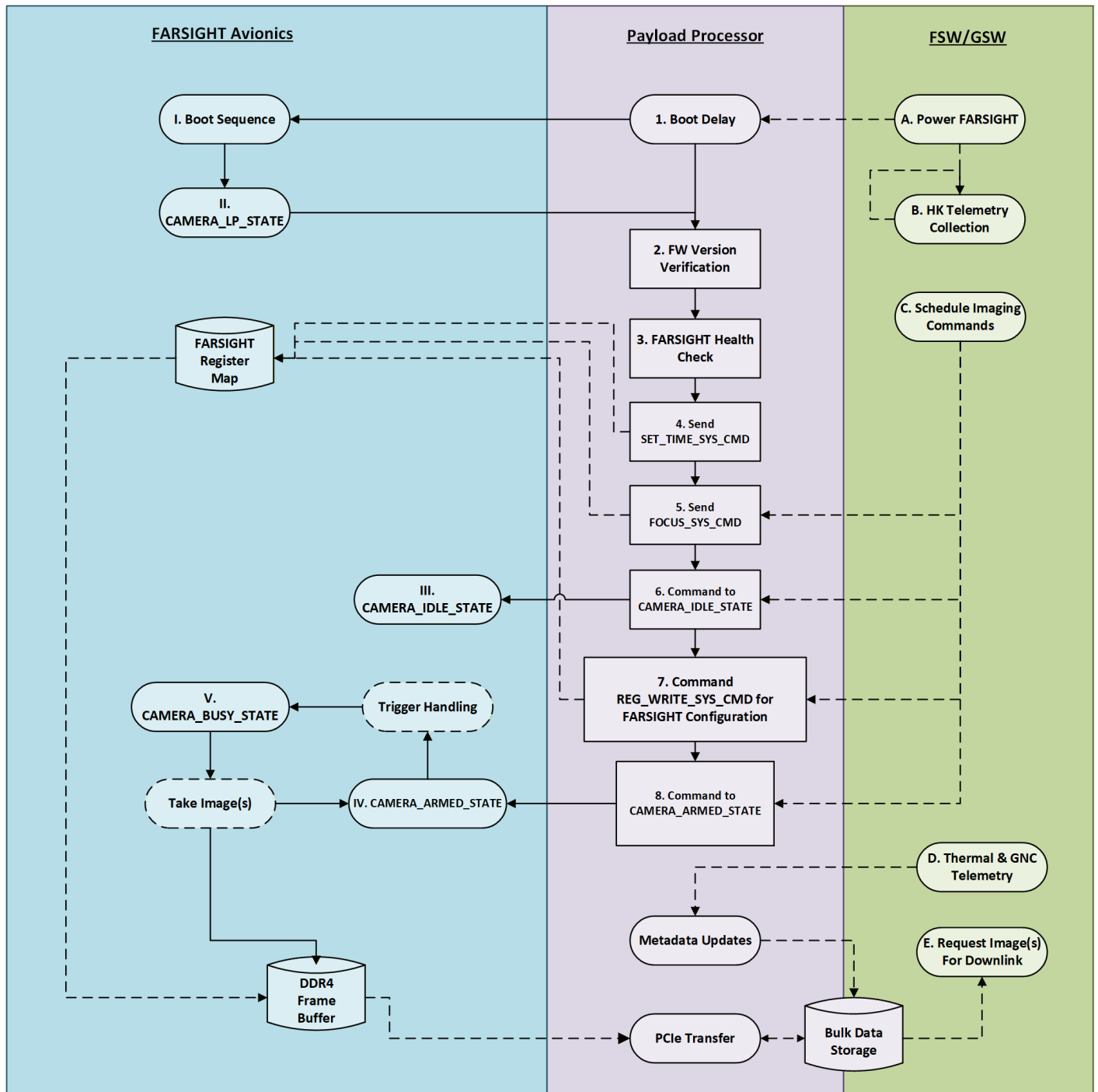


Figure 4-1: FARSIGHT Camera Operations

4.1 SYSTEM MODES

This section provides an overview of FARSIGHT’s five modes of operation. A high-level overview of these modes are provided in Table 4-1 below, with individual modes further detailed in this section. For details on mode transition commands, see the respective detailed sections for each mode.

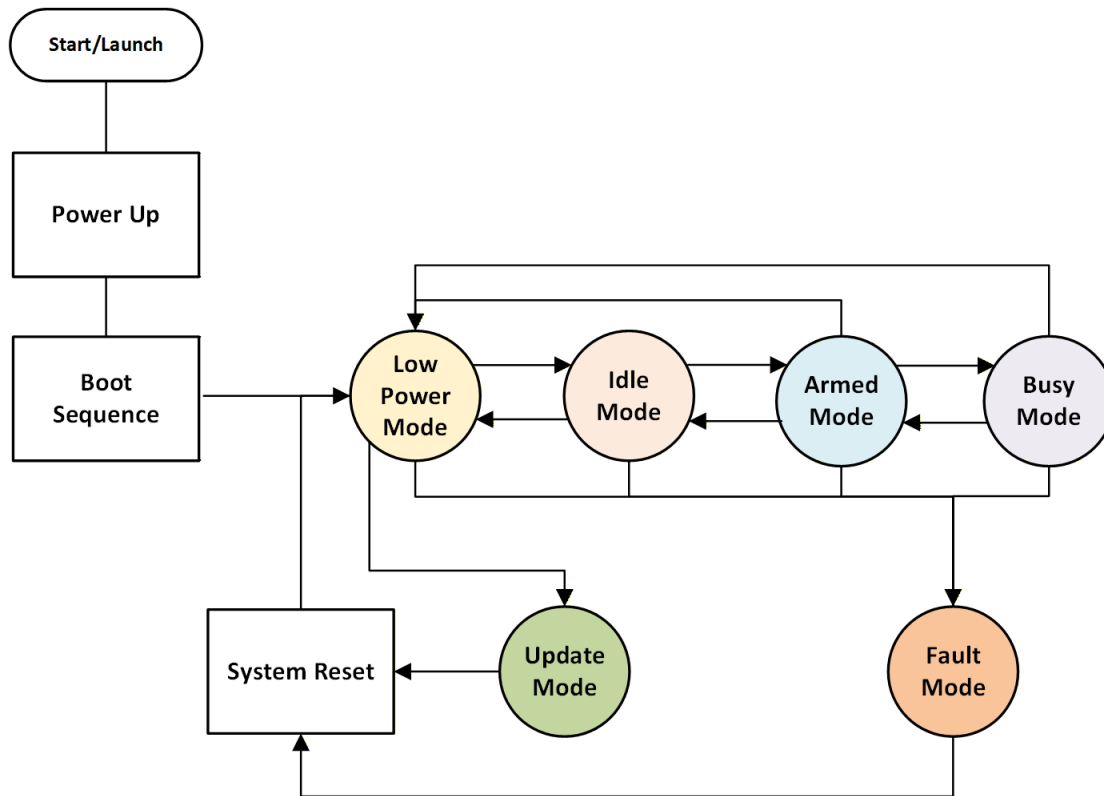


Figure 4-3: Mode Transition Diagram

Table 4-1: Modes of Operation

Mode	Description
Low Power (CAMERA_L P_STATE)	The default mode after a reset and successful boot of critical power regions. FARSIGHT can also be commanded to enter this mode. Within this mode, the camera sensor is not powered, health telemetry acquisition begins, and FARSIGHT awaits a command for mode transition. From Low Power State, FARSIGHT can be commanded to enter Idle State.
Update	Upon request, FARSIGHT will transition from Low Power Mode to Update Mode to facilitate updates of FARSIGHT’s FPGA (Field Programmable Gate Array) Firmware, RISC-V (Reduced Instruction Set Computer) software and TMTC protocol from the controlling host. While in this mode the CMOS sensor is powered off and all FPGA activity is halted other than what is required during the firmware update process. From Update Mode, FARSIGHT will automatically perform a reset or require a power cycle (To Be Refined) to then transition back into Low Power Mode.

Mode	Description
Idle (CAMERA_IDLE_STATE)	Upon request, FARSIGHT will transition from Low Power State to Idle State, where FARSIGHT will prepare for image acquisition commands and nominal payload operations. When FARSIGHT enters this state, the CMOS sensor and all required interfaces for imaging operations will be enabled to prepare for imaging commands. From Idle State, FARSIGHT can transition to Armed State or return to Low Power State.
Armed (CAMERA_ARMED_STATE)	A preparatory state where the system is configured and awaiting a capture trigger. The imaging subsystem is primed and ready for immediate operation. In this state, FARSIGHT awaits either a user-initiated cue or a scheduled capture event to begin image acquisition. From Armed State, FARSIGHT can transition to Busy State upon receiving a capture command, or return to Idle State via the disarm command.
Busy (CAMERA_BUSY_STATE)	Upon request, FARSIGHT will transition from Armed State to Busy State, where the system actively captures images and stores them into the designated DDR4 frame buffer space. Further details about the imaging operation are provided in the subsequent SDD. During imaging, certain camera sensor parameters can be updated via SPI commands; some updates will be applied immediately to the current frame, while others will take effect on the next frame. The specifics of parameter update timing are described in later sections. Upon completion of image capture, the FPGA hardware generates an interrupt to the RISC-V firmware indicating capture completion, at which point FARSIGHT transitions from Busy State back to Armed State.
Fault (CAMERA_FAULT_STATE)	A safety state entered when a housekeeping (HK) fault is detected. This state can be reached from Low Power, Idle, or Armed states when the system detects anomalous conditions that require immediate attention.

4.1.1 POWER UP & BOOT SEQUENCE

FARSIGHT is initiated by enabling the NPS channel on the host side. The FARSIGHT hardware is booted in sequence as dictated by **Table Ref.** Afterwards, FARSIGHT Avionics will check for an existing firmware image. The power region and their respective power sources will be booted one by one until the entire sequence has been successfully completed. The housekeeper FPGA will attempt to boot a respective power region up to 4 times before aborting the boot sequence.

Boot Order	Power Region	Power Sources	Has nfault?	Critical?	PolarFire SW Controlled?	Mode Enabled when	Nominal Boot Time
1	Step Down Converters	2v2	Yes	Yes	No	Boot Sequence	2.6ms
2		3V0	Yes				2ms
3		4V0	Yes				2ms
4	DDR8	2V5	Yes	No	No		0.704ms
5		1V2	Yes				0.52ms
6		0V6	No				0.4ms
7	DDR16	2V5	Yes	No	No		1.23ms
8		1V2	Yes				0.91ms
9		0V6	No				0.4ms
10	FPGA	1V0	Yes	Yes	No		0.64ms
11		1V0A	No				2.3ms
12		1V25A	No				2.3ms
13		1V8	No				2.3ms
14		1V8 IMX	No				2.3ms
15		2C5A	No				2.3ms
16		3V3 B4	No				2.3ms
17		3V3 B5	No				2.3ms
1	LVDS	3V3	No	No	Yes	2.3ms	
1	Eth1	1V0	No	No	Yes	2.3ms	
2		1V0A	No			2.3ms	
3		2V5A	No			2.3ms	
4		3V3	No			2.3ms	
1	Eth2	1V0	No	No	Yes	2.3ms	
2		1V0A	No			2.3ms	
3		2V5A	No			2.3ms	
4		3V3	No			2.3ms	
1	Stepper Primary Motor	28V0	Yes	No	Yes	Only When In Use	N/A
1	Stepper Secondary Motor	28V0	Yes	No	Yes		N/A
1	LVDT	15V0	No	No	Yes		N/A
1	IMX531	1V1	Yes	No	Yes	CAMERA_LP_STATE	1.22ms
2		1V8	No				2.3ms
3		2V9	No				0.45ms
4		3V3	No				5.1ms

Criteria Description	
Entrance Criteria	1. NPS Channel for FARSIGHT is initially disabled 2. FARSIGHT is not in any operational mode
Exit Criteria	1. Critical hardware is successfully booted 2. FPGA initialized

4.1.2 Low Power (CAMERA_LP_STATE)

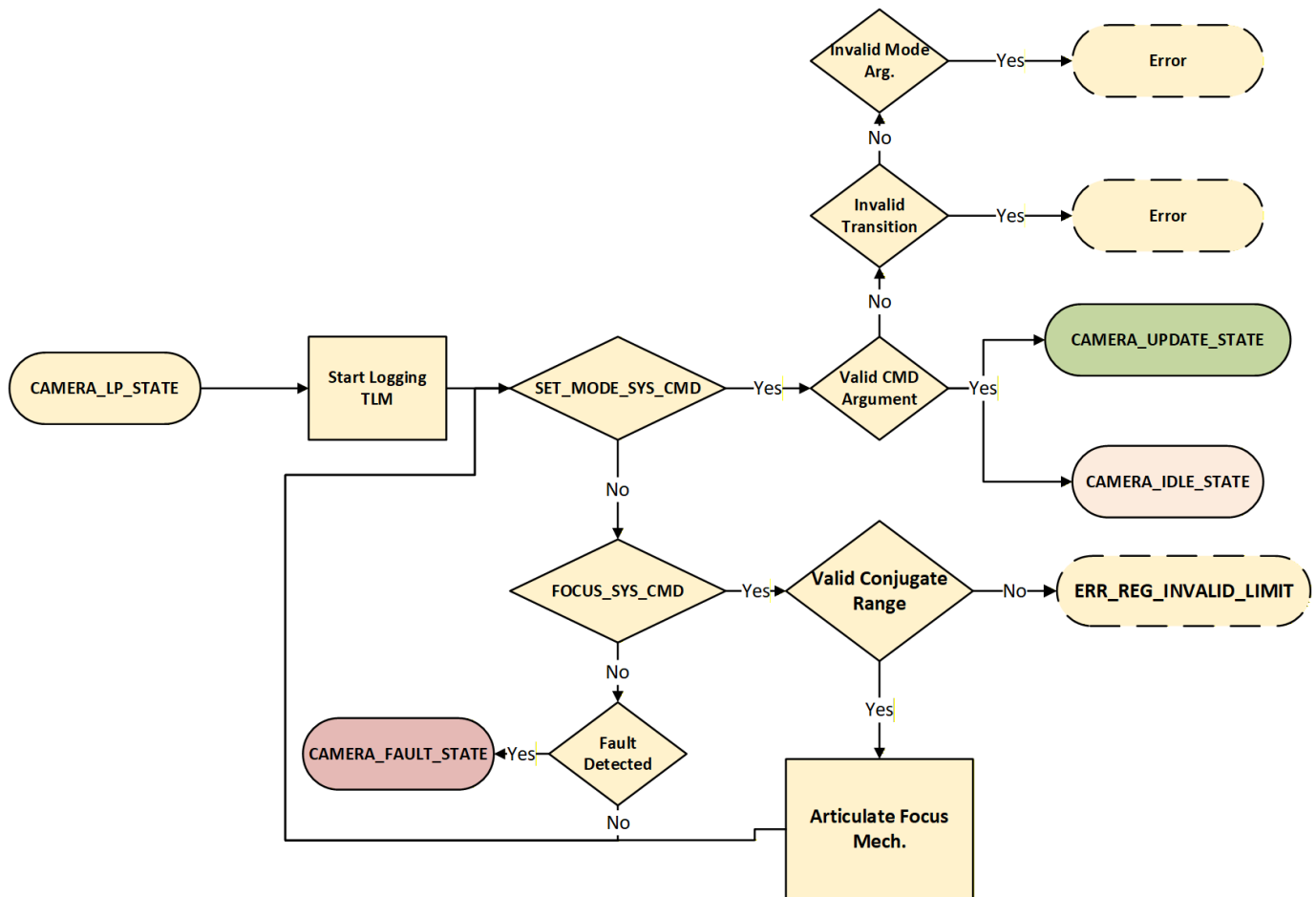
As described in Table 4-1, CAMERA_LP_STATE is the default mode FARSIGHT will enter once critical components have successfully booted or commanded to enter this mode. CAMERA_LP_STATE's entrance and exit criteria (see Table 4-2), FARSIGHT state description as well as the sequence of operations (see Figure 4-4) are described below.

After transitioning to CAMERA_LP_STATE, FARSIGHT initializes to a minimal state for communication with the host. After CAMERA_LP_STATE initialization completes, FARSIGHT will be in the state detailed in the list below:

1. CMOS Sensor is disabled
2. FPGA is powered on
3. Pins initialized to default state
4. FARSIGHT is collecting state-of-health telemetry
5. FARSIGHT is listening for commands

Table 4-3: LP_MODE-Entrance & Exit Criteria (TBR)

Criteria Description	
Entrance Criteria	3. FARSIGHT was Reset through a valid RESET_SYS_CMD command 4. Provided a valid SET_MODE_SYS_CMD command to CAMERA_LP_STATE = 0x00 5. FPGA initializes properly 6. Watchdog timer expires
Exit Criteria	3. Imaging sensor temperature within operating temperature range 4. Provided a valid SET_MODE_SYS_CMD command to either a. UPDATE_MODE = 0x01 – to initiate FARSIGHT's update process b. IDLE_MODE = 0x02– to initialize the sensor and prepare for imaging operations



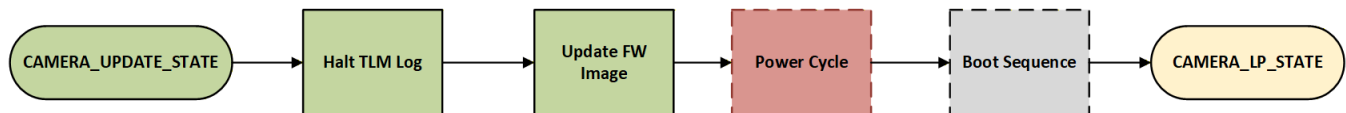
4.1.3 Update Mode (CAMERA_UPDATE_STATE):

As described in Table 4-1, FARSIGHT can be commanded to enter CAMERA_UPDATE_STATE to permit updates of firmware, software and TMTC protocol from the host. CAMERA_UPDATE_STATE's entrance and exit criteria (see Table 4-3), FARSIGHT state description as well as the sequence of operations (see Figure 4-5) are described below.

After transitioning to CAMERA_UPDATE_STATE, FARSIGHT will disable all functionality other than that which is required to update the FPGA's firmware. After the update process is completed, FARSIGHT will reset to then transition back to CAMERA_LP_STATE.

Table 4-4: UPDATE_MODE - Entrance & Exit Criteria (TBR)

	Criteria Description
Entrance Criteria	1. FARSIGHT is initially in CAMERA_LP_STATE and respective exit criteria is satisfied 2. Provided a valid SET_MODE_SYS_CMD with Opcode = 0x01
Exit Criteria	1. Host finishes transfer of firmware update 2. Valid RESET_SYS_CMD has been received



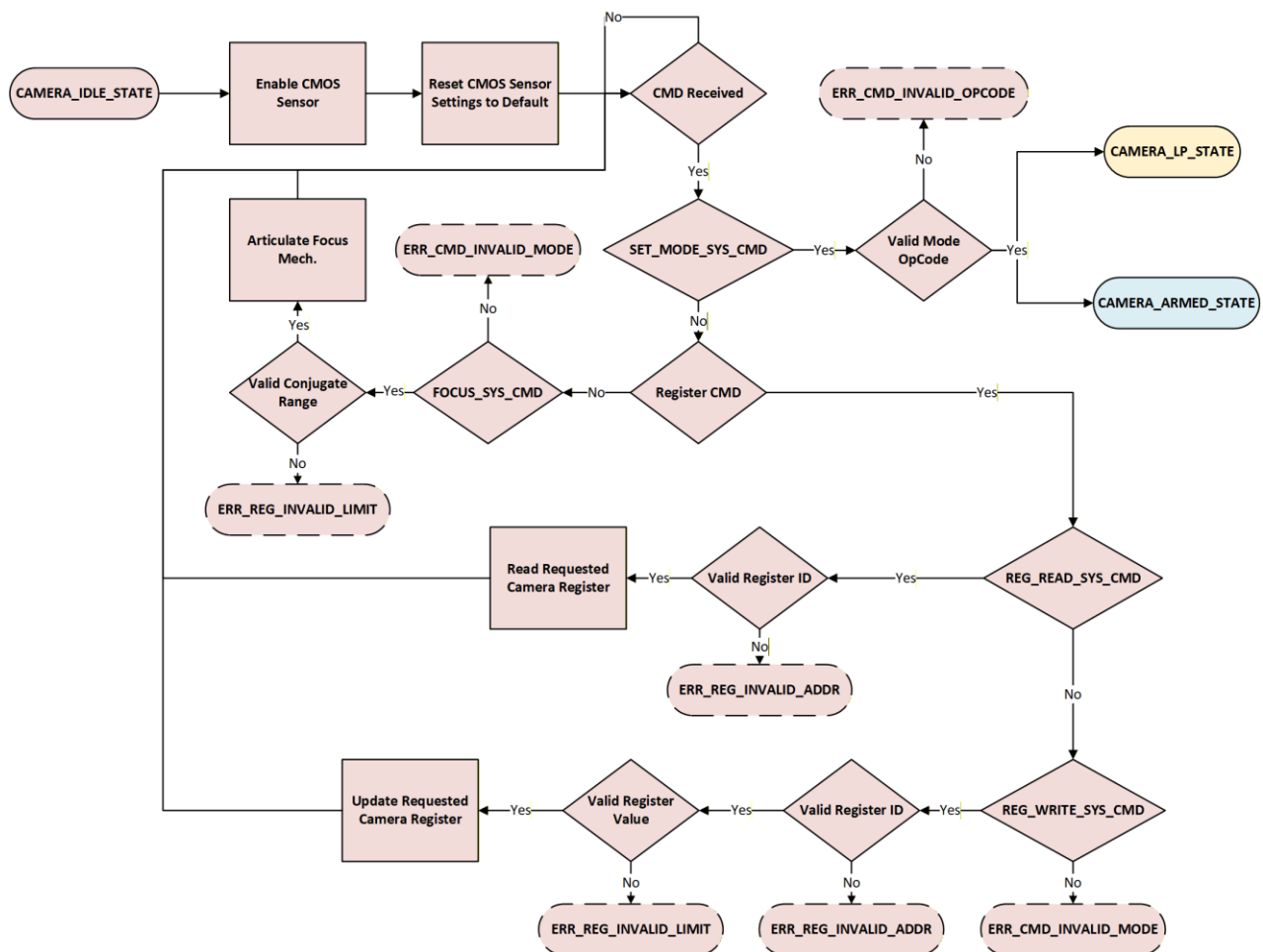
To facilitate programming, the host should use Microchip's DirectC SPI-Direct library and follow protocols as described.

4.1.4 Idle Mode (CAMERA_IDLE_STATE)

As described in Table 4-1, FARSIGHT can be commanded to enter CAMERA_IDLE_STATE to prepare for image acquisition operations. CAMERA_IDLE_STATE's entrance and exit criteria (see Table 4-4) as well as the sequence of operations (see Figure 4-6) are described below.

Table 4-5: IDLE_MODE-Entrance & Exit Criteria (TBR)

Criteria Description	
Entrance Criteria	One of the following: - FARSIGHT is in CAMERA_LP_STATE and CAMERA_LP_STATE exit criteria satisfied and provided a valid SET_MODE_SYS_CMD command to IDLE_MODE = 0x02 - FARSIGHT is in CAMERA_ARMED_STATE and CAMERA_ARMED_STATE exit criteria satisfied and provided a valid SET_MODE_SYS_CMD commanded to CAMERA_ARMED_STATE = 0xXX
Exit Criteria	One of the following: - CAMERA_LP_STATE entrance criteria satisfied and a valid SET_MODE_SYS_CMD is provided with argument CAMERA_LP_STATE = 0x00 - CAMERA_ARMED_MODE entrance criteria satisfied and a valid SET_MODE_SYS_CMD is provided with argument CAMERA_ARMED_STATE = 0x02

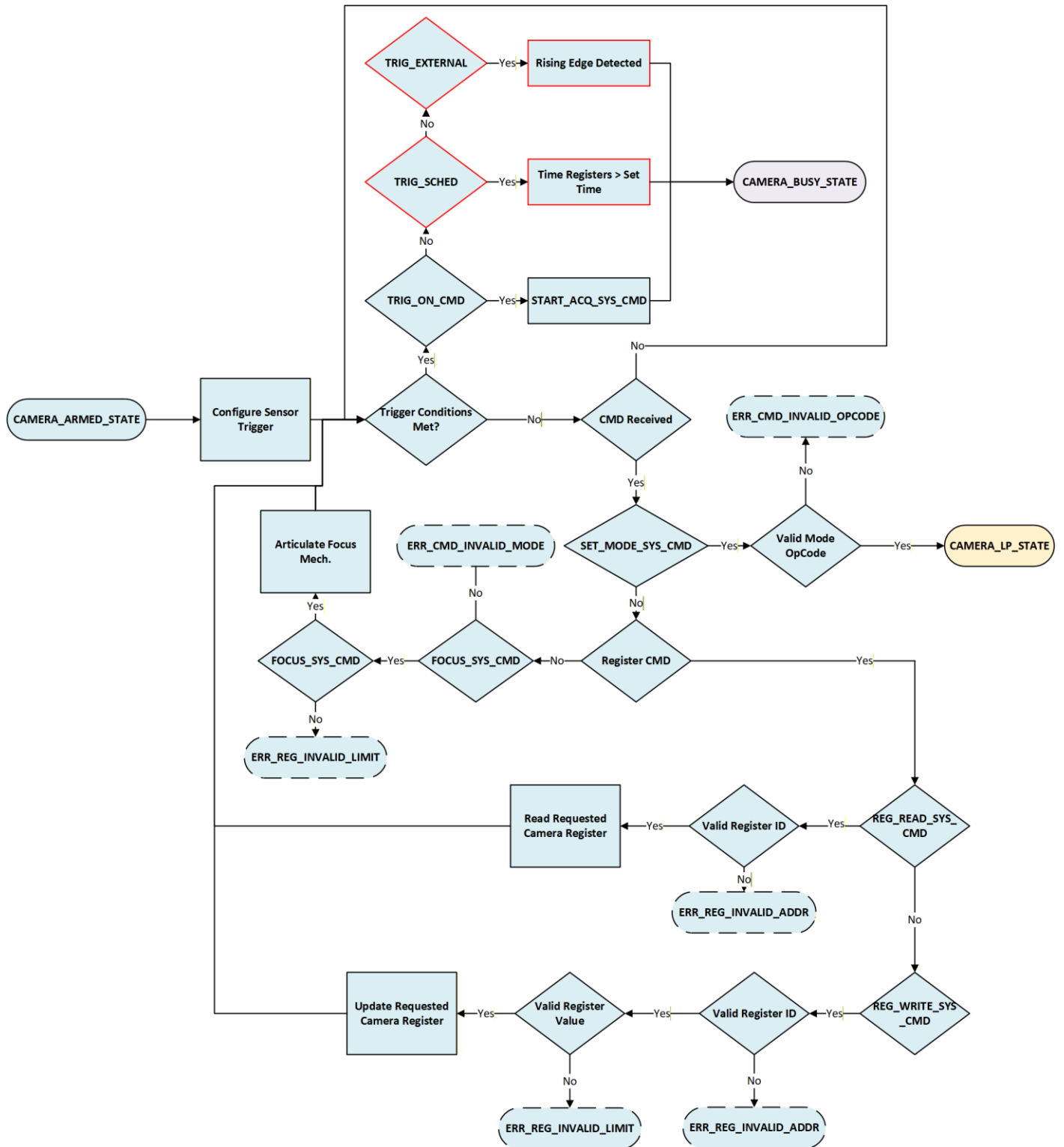


4.1.5 Armed Mode (CAMERA_ARMED_STATE)

Image acquisition starts with the initial switch to CAMERA_ARMED_STATE, where the sensor can be configured and will wait for the configured trigger to switch to CAMERA_BUSY_STATE. Registers used for imaging in CAMERA_BUSY_STATE are the registers that are set at the time of the trigger and will persist until the camera returns to CAMERA_ARMED_STATE.

Table 4-6: MANUAL_MODE – Entrance & Exit Criteria (TBR)

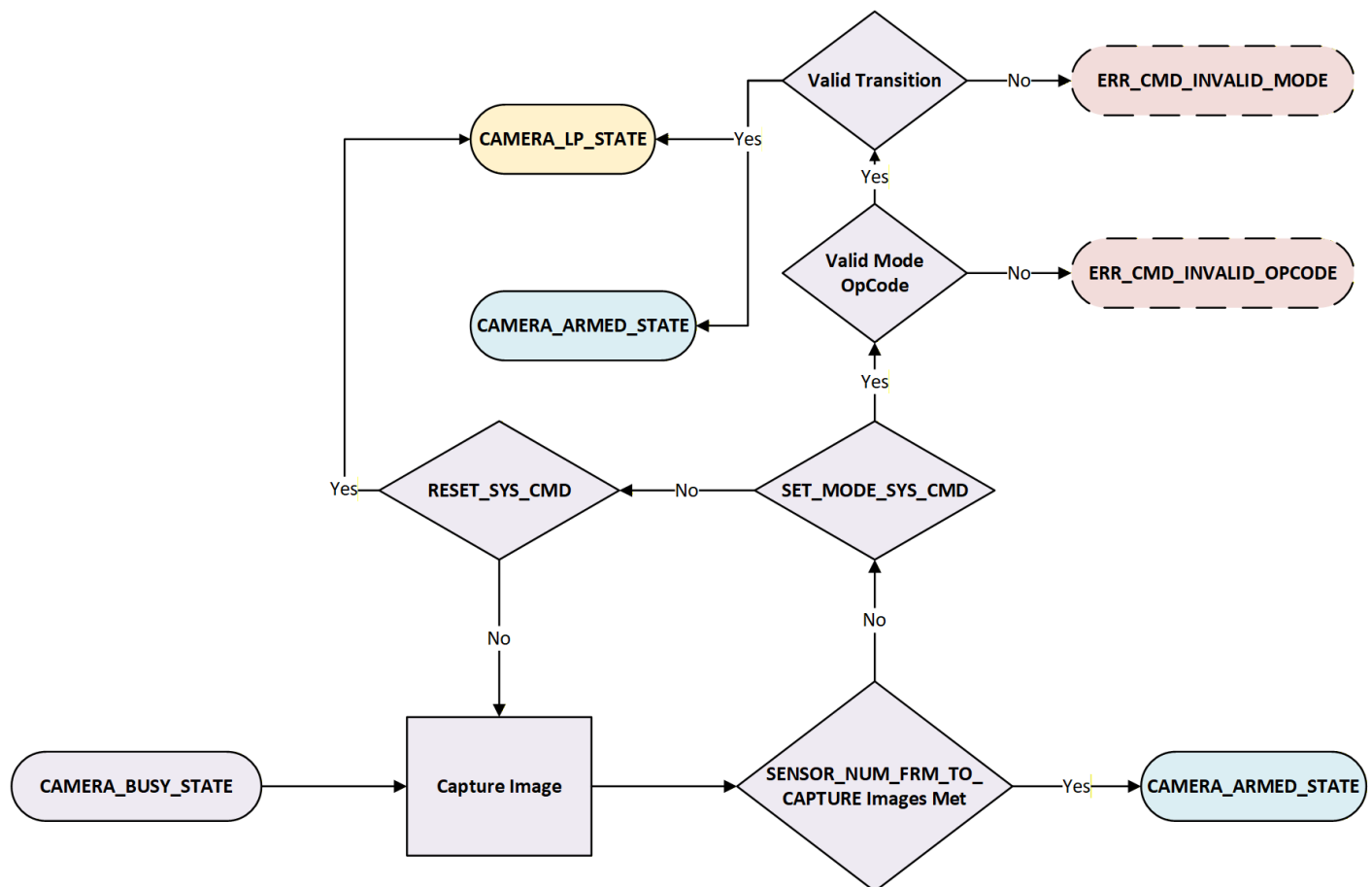
Criteria Description	
Entrance Criteria	1. FARSIGHT is in CAMERA_IDLE_STATE and exit criteria for CAMERA_IDLE_STATE have been satisfied with a valid SET_MODE_SYS_CMD with argument CAMERA_ARMED_STATE = 0x02 2. FARSIGHT is in CAMERA_BUSY_STATE and exit criteria for CAMERA_BUSY_STATE have been satisfied
Exit Criteria	1. CAMERA_LP_STATE entrance criteria satisfied and a valid SET_MODE_SYS_CMD is provided with argument CAMERA_LP_STATE = 0x00 2. CAMERA_IDLE_STATE entrance criteria satisfied and a valid SET_MODE_SYS_CMD is provided with argument CAMERA_IDLE_STATE = 0x01 3. The trigger conditions dictated by the set camera trigger in the SENSOR_TRIG_MODE_SYS_REG have been satisfied (see TABLE REF for details)



4.1.6 Busy Mode (CAMERA_BUSY_STATE)

Image acquisition with the associate camera configuration as last read during CAMERA_ARMED_STATE occurs in CAMERA_BUSY_STATE which is initiated by the conditions met with the associated set trigger in the SENSOR_TRIG_MODE_SYS_REG camera register.

Criteria Description	
Entrance Criteria	1. FARSIGHT is in CAMERA_ARMED_STATE and exit criteria for CAMERA_ARMED_STATE have been satisfied with the associated trigger conditions as dictated by the trigger mode set in the SENSOR_TRIG_MODE_SYS_REG
Exit Criteria	1. CAMERA_LP_STATE entrance criteria satisfied and a valid SET_MODE_SYS_CMD is provided with argument CAMERA_LP_STATE = 0x00 2. CAMERA_ARMED_STATE entrance criteria satisfied and a valid SET_MODE_SYS_CMD with argument CAMERA_ARMED_STATE = 0x02 3. The camera sensor captures the number of images dictated by the SENSOR_NUM_FRM_TO_CAPTURE_SYS_REG



4.1.7 Fault Mode (CAMERA_FAULT_STATE)

If a critical power region in the FAV hardware that has an nfault signal or a critical pgood signal as described by **TABLE REF** goes low for at least 5 microseconds, the FARSIGHT system will repeatedly send a UART telemetry message through the RS422 lane. The structure of that message can be seen in **TABLE REF**.

UART Message Format			
Power Region	Byte #	Bitfield	Value/Source Status
N/A	0	7:0	0xDE
N/A	1	7:0	0xAD
N/A	2	7:0	0xBE
N/A	3	7:0	0xEF
FPGA	4	7	3V3 B5 pgood
		6	3V3 B4 pgood
		5	2V5A pgood
		4	1V8 IMX pgood
		3	1V8 pgood
		2	1V25A pgood
		1	1V0A pgood
		0	1V0 pgood
FPGA	5	7:1	Reserved
		0	1V0 nfault
LVDS	6	7:1	Reserved
		0	3V3 pgood
DDR16	7	7:5	Reserved
		4	0V6 pgood
		3	1V2 pgood
		2	2V5 pgood
		1	1V2 nfault
		0	2V5 nfault
DDR8	8	7:5	Reserved
		4	0V6 pgood
		3	1V2 pgood
		2	2V5 pgood
		1	1V2 nfault
		0	2V5 nfault
Step Down Converters	9	7:6	Reserved
		5	4V0 pgood
		4	3V0 pgood
		3	2V2 pgood
		2	4V0 nfault
		1	3V0 nfault
		0	2V2 nfault
N/A	10	7:0	0xBA

N/A	11	7:0	0xDD
N/A	12	7:0	0xFE
N/A	13	7:0	0xED

In the event of a non-critical failure, housekeeper status can be monitored through a REG_READ_SYS_CMD of the HOUSEKEEPER_STATUS_SYS_REG system register, with bitfields corresponding to individual power regions (SEC REF).

Software failures can be detected via the error outputs seen in **SEC REF**. In the event of a software failure, an update to the firmware may be required.

4.2 NOMINAL IMAGING OPERATIONS

In this section we will discuss FARSIGHT imaging operation flow.

The imaging operation flow consists of the following parts:

- Initialization and Health Check
- Camera Configuration
- Focus adjustment
- Image acquisition and offload

IT TAKES AT LEAST 6.8 MILLISECONDS AFTER TRANSITION TO IMAGING MODES TO BEGIN IMAGING. THIS NEEDS TO BE FURTHER CHARACTERIZED THROUGH AVIONICS BOARD TESTING TO PROVIDE A FINAL VALUE.***

4.2.1 Initialization

Before beginning image acquisition, verify operation and health of the FARSIGHT system. Before switching the FARSIGHT camera to CAMERA_IDLE_STATE, verify that the camera is currently in CAMERA_LP_STATE via the REG_READ_SYS_CMD command (reading the SENSOR_STATE_SYS_REG). Also verify the appropriate firmware version is being used via the REG_READ_SYS_CMD command (reading the VERSION_SYS_REG). Send an ECHO_SYS_CMD to verify the return value is what is expected. Complete the functionality check by performing a REG_WRITE_SYS_CMD command to the NOOP_SYS_REG and reading back the register to confirm read/write functionality.

Before switching to idle mode, verify functionality of the DDR4 and ethernet components via a REG_READ_SYS_CMD of the relevant PWRcamera register. Verify bits:

4.2.2 Camera Configuration

Switch the camera to CAMERA_IDLE_STATE via the SET_MODE_SYS_CMD (Op Code = 0x01). During this state, the FARSIGHT Camera Registers can be updated and/or read back to the host. Registers can also be updated in CAMERA_ARMED_STATE and will be used for image capture incident on the next trigger. While registers can be updated while in CAMERA_BUSY_STATE, they will not be recognized as imaging parameters until the SENSOR_NUM_FRM_TO_CAPTURE_SYS_REG value is satisfied and the sensor returns to CAMERA_ARMED_STATE. The protocols for updating both currently implemented writeable camera, system, and network registers can be seen in the tables below:

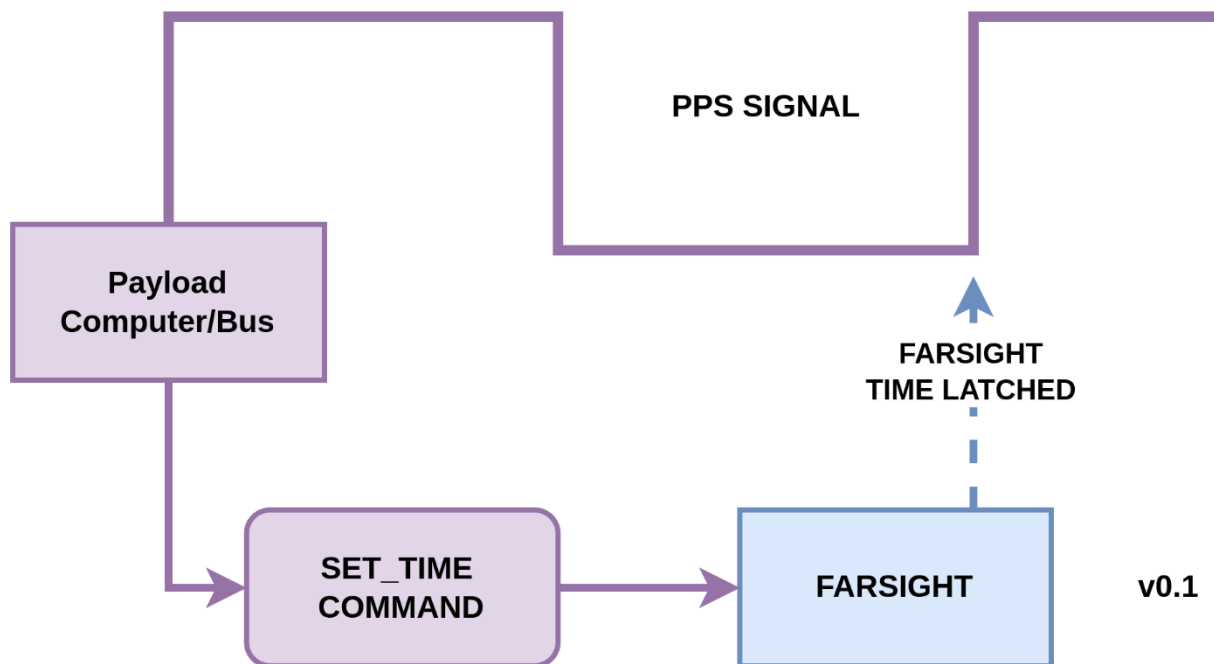
CAMERA REGISTER	TAKES AFFECT	ACCEPTED MODES	RELEVANT COMMAND
SENSOR_FRM_MOD_SYS_REG	Next Trigger	CAMERA_IDLE_STATE CAMERA_ARMED_STATE	REG_WRITE_SYS_CMD
SENSOR_BIT_DEPTH_SYS_REG			
SENSOR_GAIN_SYS_REG			
SENSOR_BLO_SYS_REG			
SENSOR_EXPO_USEC_SYS_REG			
SENSOR_OVER_TEMP_KEL_SYS_REG			

SYSTEM REGISTER	TAKES AFFECT	ACCEPTED MODES	RELEVANT COMMAND
TRIG_MODE_SYS_REG	Next Trigger	CAMERA_IDLE_STATE CAMERA_ARMED_STATE	REG_WRITE_SYS_CMD
IMG_PER_TRIGGER_SYS_REG			
FRAMES_PER_SEC_SYS_REG			
CAP_START_DLY_USEC_SYS_REG			
FOCUS_REQ_DIST_M_SYS_REG	Reception of Valid FOCUS_SYS_CMD	CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_BUSY_STATE	FOCUS_SYS_CMD
TIMESTAMP_UNIX_SYS_REG	Next PPS Rising Edge	CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_BUSY_STATE	SET_TIME_SYS_CMD
GPO_CONF_SYS_REG	Next Trigger	CAMERA_IDLE_STATE CAMERA_ARMED_STATE	REG_WRITE_SYS_CMD
GPI_CONF_SYS_REG			
INTR_PIN_CONF_SYS_REG			
TRIG_PIN_CONF_SYS_REG			
TRIG_DELAY_USEC_SYS_REG			
TRIG_TIMEOUT_MSEC_SYS_REG			
TMTC_TIMEOUT_MSEC_SYS_REG			
SRC_MAC_ADDRESS_0_3_SYS_REG			
SRC_MAC_ADDRESS_4_5_SYS_REG			
DST_MAC_ADDRESS_0_3_SYS_REG			
DST_MAC_ADDRESS_4_5_SYS_REG			
SRC_IP_ADDRESS_SYS_REG			
DEST_IP_ADDRESS_SYS_REG			

GATEWAY_SYS_REG			
SUBNET_MASK_SYS_REG			
SRC_PORT_SYS_REG			
DST_PORT_SYS_REG			

FARSIGHT's time should be set as a part of the initialization process to ensure that the registered time matches. The SET_TIME_SYS_CMD command will provide FARSIGHT a real time as detailed in the table below and latch this value on the next rising edge of the PPS signal provided to FARSIGHT.

Field	Size (bits)	Bit Range	Description
TIMESTAMP_UNIX_EPOCH_SEC	32	63:32	Seconds since the epoch (32-bit signed integer, supporting 136 years from the epoch January 1, 1970, for Unix)
TIMESTAMP_SUBSEC_NSEC	32	31:0	Sub-seconds component of the UNIX time since epoch in nanoseconds



After an initial SET_TIME_SYS_CMD, FARSIGHT will automatically set the TIMESTAMP_UNIX_EPOCH_SEC_SYS and TIMESTAMP_SUBSEC_NSEC_SYS_REG. Depending on integrity of the PPS signal, this setting may or may not reflect the real system time. To mitigate this risk, use a SET_TIME_SYS_CMD before image acquisition.

After ethernet configuration, ping the ETH1 interface to verify functionality.

Before image acquisition, make sure that the camera has been transitioned to CAMERA_ARMED_STATE via a valid SET_MODE_SYS_CMD.

4.2.3 Focusing (Optional)

To adjust the optimal focus point of the FARSIGHT sensor, send a FOCUS_SYS_CMD with the requested target range in meters. This will update the FOCUS_REQ_DIST_M_SYS_REG in the system registers. The FARSIGHT focus mechanism articulates in response to a valid command. The focus mechanism takes approximately 2 seconds to travel the entire throw of the focus mechanism. Verification of the articulation of the focus mechanism can be done via a REG_READ_SYS_CMD of the FOCUS_COMP_DIST_M_SYS_REG or FOCUS_LVDT_POS_NM_REG registers. Allow the focus mechanism ample time to reach its focus point before initiating image acquisition. The focus mechanism can be articulated while the camera is in any mode outside of the CAMERA_FAULT_STATE mode.

4.2.4 Image Acquisition and Offload

Image acquisition begins after the associated trigger conditions dictated by the SENSOR_TRIG_MODE_SYS_REG. The camera sensor will then enter CAMERA_BUSY_STATE and capture images using the settings dictated by the associated register values at the time of trigger recognition. The camera will capture the number of images dictated by the SENSOR_NUM_FRM_TO_CAPTURE_SYS_REG and store them to the DDR4 frame buffer starting with the index set by the BUFF_WRITE_INDEX_SYS_REG. After finishing image acquisition, the camera will automatically return back to CAMERA_STATE_ARMED and await the next trigger. The camera can also be returned back to CAMERA_LP_STATE or CAMERA_IDLE_STATE via a valid SET_MODE_SYS_CMD.

To offload images off the frame buffer, send an XFER_BUFF_SYS_CMD command with the associated image index and number of frames.

4.3 PPS TIMEKEEPING:

The external 1PPS signal synchronizes the real system time to the internal FARSIGHT time. This can be explicitly called for with a SET_TIME_SYS_CMD but will be automatic with the host to FAV 1PPS. The FARSIGHT internal clock has a frequency of 50 MHz and will handle internal time keeping in reference to the received 1PPS pulse signal. An 8-sample moving average is used to calculate the average period of input for the 1PPS. This period is used to verify whether an input 1PPS is ignored.

Each time a 1PPS comes in, the duration since the last 1PPS pulse is checked. If the duration since last PPS input is <50% or >150% of the expected 1 PPS period governed by the tick count since last 1PPS signal, that signal is then ignored and will not affect local timekeeping registers.

It is recommended that a SET_TIME_SYS_CMD proceeds all imaging activities to mitigate potential errors.

4.4 RESETS & WATCHDOGS:

Resets of the FARSIGHT system are either initiated with a valid RESET_SYS_CMD or by initiating a power cycle. The valid resets are:

- **Camera Sensor Reset (RESET_SYS_CMD – Bit 0):** Reset of the camera sensor into CAMERA_LP_STATE
- **Focus Mechanism Reset (RESET_SYS_CMD – Bit 1):** Reset all focus mechanism faults and stop driving the motor controllers at their current position
- **Register Reset (RESET_SYS_CMD – Bit 2):** Reset the camera registers to their default values
- **Power Cycle:** Cut power to the NPS Channel and reenable to initiate a power cycle, which will re-initialize the boot sequence

RISCV Processor Watchdog: In case of RISCV processor fatal exception the core will be captured in the trap handler until the core is reset. A watchdog timer is implemented in the PolarFire FPGA to watch for Software hangs and reset the RISCV core. Upon Software reset the application will initialize all the Software registers and will place the IMX camera into a known state. The details of WDT and after reset status capture is included in SDD. *Future Implementation

4.4.1

4.5 LATCHUP & BOOT FAILURE RESPONSE:

When latchups from radiation events or components fail to boot, FARSIGHT responds depending on the criticality and circumstance of the failure. There are two primary responses:

FARSIGHT enters (or stays in) BOOT_MODE and beacons a statically sized UART packet that provides information on state/failure of critical power regions or DDR4 if a latchup occurs.

This occurs when a critical power region fails to boot OR latches up

This occurs when DDR4 latches up but DOES NOT occur if DDR4 fails to boot. During the boot process, failures will be reported on DDR, but the boot process will continue since it is not a critical memory section.

When this response occurs, a Power on Reset is required to rectify the error.

FARSIGHT disables power regions with higher boot orders than the failed power region as long as this region is a PolarFire SW controlled region as defined in the table below.

When this response occurs, it is recommended to power cycle FARSIGHT after requesting telemetry.

These failures can be detected by regularly checking the relevant FARSIGHT system registers

5.0 SOFTWARE TMTC INTERFACE

In this section we will define the Telemetry and Telecommand (TMTC) interface between the host computer (HOST) and FARSIGHT Avionics computer (FAV). Before going over the details let's go over some terminologies used in this section.

Table 5-1: Software TMTC Interface Terminologies

Term	Description
Host	Host is the computer that communicates with the FARSIGHT Avionics Computer by sending telecommands (TC) and receiving telemetry (TL).
TMTC Packet	Telemetry and Telecommand packet is a formatted unit of data that is transmitted between the Host and FAV. TMTC Packet consists of a Header and a Datagram.
Datagram	Datagram or payload is the actual information or message being sent between FAV and Host. A Datagram can contain its own header format plus payload data or can only contain the payload data without any extra header.
Command	Commands are a set of instructions sent from the host to the FARSIGHT Avionics computer to control operations and configuration of the FARSIGHT payload.
Science Payload	FARSIGHT science data payload are images captured by the FARSIGHT camera sensor.

5.1 SERIAL INTERFACE

A quick reference for FARSIGHT's serial debug and backup TMTC interface is provided in the table below.

Table 5-2: FARSIGHT's Serial Debug and Backup TMTC Interface Quick Reference

Parameter	Value
Stop Bits	1
Parity	None
Data size	8-bits
Baud Rate	115200 (default, but configurable)

5.2 TMTC PACKET FORMAT

FARSIGHT has a custom packetization protocol whose packet format is defined in Figure 5-1. This protocol operates in a synchronous point-to-point request-reply pattern in which the host (computer interfacing with FARSIGHT) will send a command and then receive a response from FARSIGHT Avionics Computer (FAV) based on that command. This packet information is always sent from the most significant byte (MSB) first.

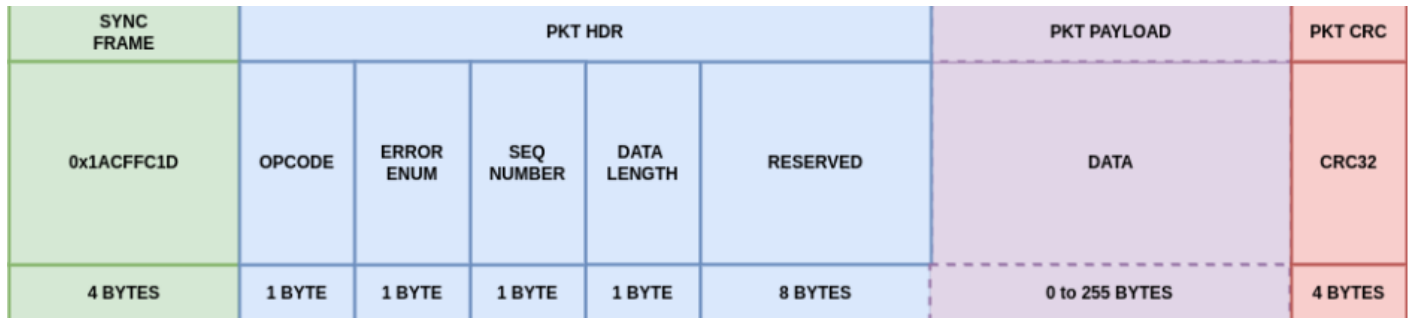


Figure 5-1:TMTC Packet Format

The CRC-32 is calculated over all bytes except sync frame for data integrity verification.

The FARSIGHT TMTC protocol uses mixed byte ordering. All packet framing and header fields (sync frame, reserved, and CRC) are transmitted in big-endian (most significant byte first) order. Single-byte fields (opcode, error code, sequence number, and data length) have no byte-order dependency. The data payload is transmitted in **little-endian** (least significant byte first) order, consistent with the native byte order of the RISC-V processor on the FAV.

5.2.1 PKT_HDR

Total Size: 16 bytes

Description: This packet header provided the FAV and host with sufficient information to decode commands and telemetry packets. See Table 5-3 for a breakdown of each field within this header.

Field	Size (bits)	Description	Byte Order
SYNC FRAME	32	Used to indicate the start of a packet. SYNC FRAME FOR PACKET WILL ALWAYS BE 0x1ACFFC1D (Transmitted as 0x1A, 0xCF, 0xFC, 0x1D)	Big-endian
OPCODE	8	Command or function code identifying the operation	-
ERROR ENUM	8	Error status (used in reply packets from FAV)	-
SEQ NUMBER	8	Packet sequence counter for tracking purposes	-

DATA LENGTH	8	Payload data length in bytes	-
Reserved	64	Reserved for future functionality implementations	Big-endian

Table 5-3: Packet Header Field Descriptions

5.2.2 PKT_DATA

Field with a maximum size of 255 bytes including data described by the TMTC packet header. When packet is sent from the Host to the FAV this field will include command parameters, the format of which is described for each command in section **5.3 FARSIGHT COMMANDS**. When a packet is sent from the FAV to the Host in response to a **REG_READ_SYS_CMD** (see section **5.3.2.4 REG_READ_SYS_CMD**), then data will be formatted as described in the representative register definition seen in **sec ref – sec ref**. For multi-argument commands, the FAV expects data ordered by the argument numbers given in **SEC REF – SEC REF**. An example can be seen below for a **REG_WRITE_SYS_CMD**.

PLACEHOLDER

5.2.3 TL DATAGRAM FORMAT

When the packet is sent from the FAV to the Host in response to a **REG_READ_SYS_CMD** a TL DATAGRAM will be provided as seen in the figure below. Currently, all registers are 32-bit data packets of one or multiple fields.

TL Datagram Identifier HDR			FARSIGHT USER REGISTER DATA
START_REG_ADDR (start address)	DATA_SIZE_BYTE (number of bytes read)	TL Timestamp	
4 BYTES	2 BYTES	8 BYTES	0 to 241 BYTES

Figure 5-2: TL Event, TL Channel & Raw Image Segment Datagram Formatting

5.3 FARSIGHT COMMANDS

FARSIGHT's commands are defined within this section. Many of these commands will lead to specific operations guided by the register map defined in Section 5.5 FARSIGHT REGISTERS. HIGHLIGHTED = NOT IMPLEMENTED YET.

COMMAND	OPCODE	DESCRIPTION
GET_VERSION_SYS_CMD	0x00	Get HW and SW version Info
SET_MODE_SYS_CMD	0x01	Put camera in indicated mode per section 4.1
RESET_SYS_CMD	0x02	Resets the FARSIGHT subsystems
REG_READ_SYS_CMD	0x03	Reads a specified register value
REG_WRITE_SYS_CMD	0x04	Writes a value to a specified register
SET_TIME_SYS_CMD	0x05	Sets FARSIGHT's absolute system time
START_ACQ_SYS_CMD	0x06	Starts an image acquisition
STOP_ACQ_SYS_CMD	0x07	Stops an image acquisition
FOCUS_SYS_CMD	0x08	Initiates a focus operation
HALT_FOCUS_SYS_CMD	0x09	Stop focus operation
XFER_BUFF_SYS_CMD	0x0A	Transfers a section of the DDR4 frame buffer
RESET_BUFF_SYS_CMD	0x0B	Update focus mechanism calibration table
GET_INFO_SYS_CMD	0x0C	Get system information in regard to register allowances
CONF_EVENT_LOG_SYS_CMD	0x0D	Configure the event log.
EN_TL_CHAN_LOG_SYS_CMD	0x0E	Command to configure LOG buffer.
XFER_LOG_SYS_CMD	0x0F	Transfer LOG from FAV to host
ZERO_FUNCTION	0x1E	Initiates a focus operation to the null section (0 microns) of LVDT position sensor.
SET_POSITION	0x1C	Initiates a focus operation to some position within the LVDT measurement range of -1.2mm to 1.2mm in microns.

5.3.1 Error Enumerations

ERROR	ENUM	DESCRIPTION
NO_ERROR	0x00	No error has occurred
ERR_CMD_INVALID_MODE	0x01	A command was rejected due to the current mode of the payload
ERR_CMD_INVALID_OPCODE	0x02	Command was determined to be invalid based on opcode
ERR_PKT_INVALID_HDR_CRC	0x03	Packet HEADER_CRC failed validation
ERR_PKT_INVALID_DATA_CRC	0x04	Packet DATA_CRC failed validation for commands with arguments
ERR_PKT_INVALID_DATA_LEN	0x05	Packet USER_DATA_LENGTH field was invalid

ERR_PKT_INVALID_HDR_PARAM	0x06	Invalid packet header parameter value
ERR_REG_INVALID_ADDR	0x07	Register address was invalid
ERR_REG_INVALID_LIMIT	0x08	Parameter update was determined invalid due to the value's limits
ERR_REG_WRITE_DENIED_RO	0x09	The telemetry value was read only
ERR_REG_MODE_REJECTED	0x0A	A register write was rejected due to the current mode of the payload
ERR_TBD	...	Additional error codes are expected to be added

5.3.2 Command Descriptions

Command descriptions are tabulated throughout this section. Error responses specific to the command are also provided. Although not documented in these tables, common errors are listed below:

INVALID_HDR_CRC error – returned if the HEADER_CRC field does not match the CRC computed by FARSIGHT on the header

INVALID_DATA_CRC error – returned if the DATA_CRC field does not match the CRC computed by FARSIGHT on the data field for commands that include data here

INVALID_CMD error – the opcode provided does not exist in FARSIGHT’s command table

5.3.2.1 GET_VERSION_SYS_CMD

NAME	VALUE
Description	This command will get information on the current HW and SW versions of the FARSIGHT system.
Opcode	0x00
Accepted Modes	CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_BUSY_STATE
Invalid Modes	CAMERA_FAULT_STATE
Arguments	No unique arguments identified
Return Value	TBD
Command Specific Errors	PLACEHOLDER

5.3.2.2 SET_MODE_SYS_CMD

NAME	VALUE
Description	This command will transition the camera sensor to various operational states.
Opcode	0x01
Accepted Modes	CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_BUSY_STATE
Invalid Modes	CAMERA_FAULT_STATE
Argument 1	Parameter Name: MODE
	Type: uint8_t
	Length (bytes): 1
	Description: Enumeration of different camera states: • 0x01: CAMERA_LP_STATE

		0x02: CAMERA_IDLE_STATE 0x04: CAMERA_ARMED_STATE 0x08: CAMERA_UPDATE_STATE
	Valid Values:	0x01, 0x02, 0x04, 0x08
Return Value	None	
Command Specific Errors	TBD	

5.3.2.3 RESET_SYS_CMD

NAME	VALUE	
Description	This command will perform a reset of various FARSIGHT subsystems	
Opcode	0x02	
Accepted Modes	CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_BUSY_STATE	
Invalid Modes	CAMERA_FAULT_STATE	
Argument 1	Parameter Name:	target_reset
	Type:	uint8_t
	Length (bytes):	1
	Description:	Bit field to specify the target subsystem to be reset: - Bit 0: RESET_CAM_SENSOR, setting this bit will put the sensor in the Low Power Mode and reset registers to default values - Bit 1: RESET_FOCUS_MECH, setting this bit will reset all the focus mechanism faults and stop driving the motor controllers at their current position. - Bit 2: RESET_REG_SPACE, setting this bit will reset all the firmware registers to their default values. - Bit 3: RESET_MEM_BUFF, setting this bit will reset the image frame memory buffer to index. - Bit 7-4: Not used.
	Valid Values:	0x00 to 0x0F
Return Value	None	
Command Specific Errors	TBD	

5.3.2.4 REG_READ_SYS_CMD

NAME	VALUE
------	-------

Description	This command will read a specified register value.	
Opcode	0x03	
Accepted Modes	CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_BUSY_STATE CAMERA_FAULT_STATE	
Invalid Modes	N/A	
Argument 1	Parameter Name:	reg_id
	Type:	uint8_t
	Length (bytes):	1
	Description:	Register ID to read
	Valid Values:	0x00 to 0xFF
Return Value	uint32_t	
Command Specific Errors	PLACEHOLDER	

5.3.2.5 REG_WRITE_SYS_CMD

NAME	VALUE	
Description	This command will update a specified register value.	
Opcode	0x04	
Accepted Modes	CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_BUSY_STATE	
Invalid Modes	CAMERA_LP_STATE CAMERA_FAULT_STATE	
Argument 1	Parameter Name:	REG_ID
	Type:	uint8_t
	Length (bytes):	1
	Description:	Register ID to read
	Valid Values:	0x00 to 0xFF
Argument 2	Parameter Name:	REG_VALUE
	Type:	uint32_t
	Length (bytes):	4
	Description:	Register value
	Valid Values:	Varied based on the register. See SEC REF for more info.

Return Value	None
Command Specific Errors	PLACEHOLDER

5.3.2.6 SET_TIME_SYS_CMD

NAME	VALUE	
Description	After receiving this command, the time stamp of FARSIGHT will be set on the next rising edge of the PPS signal	
Opcode	0x05	
Accepted Modes	CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_BUSY_STATE	
Invalid Modes	CAMERA_FAULT_STATE	
Argument 1	Parameter Name:	TIMESTAMP_UNIX_EPOCH_SEC
	Type:	uint32_t
	Length (bytes):	4
	Description:	Whole seconds of the UTC timestamp express as the number of seconds elapsed since the UNIX epoch (January 1, 1970 00:00:00 UTC). This value is latched on the next rising edge of the PPS signal.
	Valid Values:	0x0000_0000 to 0xFFFF_FFFF
Argument 2	Parameter Name:	TIMESTAMP_SUBSEC_NSEC
	Type:	uint32_t
	Length (bytes):	4
	Description:	Sub-second portion of the UTC timestamp in nanoseconds. Combined with TIMESTAMP_UNIX_EPOCH_SEC, this provides microsecond resolution time synchronization. This value is latched on the next rising edge of the PPS signal.
	Valid Values:	Varied based on the register. See SEC REF for more info.
Return Value	None	
Command Specific Errors	PLACEHOLDER	

5.3.2.7 START_ACQ_SYS_CMD

NAME	VALUE
Description	Requests FARSIGHT to take pictures upon receiving this command. This command will only take effect if register SENSOR_TRIG_MOD_SYS_REG is set to TRIG_ON_CMD value. For other trigger configurations an error will be returned.
Opcode	0x06

Accepted Modes	CAMERA_ARMED_STATE
Invalid Modes	CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_BUSY_STATE CAMERA_FAULT_STATE
Arguments	No unique arguments identified
Return Value	None
Command Specific Errors	PLACEHOLDER

5.3.2.8 STOP_ACQ_SYS_CMD

NAME	VALUE
Description	Requests FARSIGHT to halt image acquisition effective immediately and return the camera to the CAMERA_LP_STATE .
Opcode	0x07
Accepted Modes	CAMERA_BUSY_STATE
Invalid Modes	CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_FAULT_STATE
Arguments	No unique arguments identified
Return Value	None
Command Specific Errors	PLACEHOLDER

5.3.2.9 FOCUS_SYS_CMD

NAME	VALUE
Description	Calculates a focus distance and commands motion to that position.
Opcode	0x08
Accepted Modes	CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_BUSY_STATE
Invalid Modes	CAMERA_FAULT_STATE
Argument 1	Parameter Name: RAW_POS
	Type: uint32_t
	Length (bytes): 4

	Description:	Target conjugate distance in m
	Valid Values:	5000 - 10000000
Return Value	TBD	
Command Specific Errors	PLACEHOLDER	

5.3.2.10HALT_FOCUS_SYS_CMD

NAME	VALUE
Description	Requests FARSIGHT to halt focusing commands effective immediately.
Opcode	0x09
Accepted Modes	CAMERA_BUSY_STATE CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_ARMED_STATE
Invalid Modes	CAMERA_FAULT_STATE
Arguments	No unique arguments identified
Return Value	None
Command Specific Errors	PLACEHOLDER

5.3.2.11XFER_BUFF_SYS_CMD

NAME	VALUE	
Description	This command initiates the transfer of files from the DDR4 frame buffer to the bulk data storage aboard the spacecraft.	
Opcode	0x0A	
Accepted Modes	CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_ARMED_STATE	
Invalid Modes	CAMERA_BUSY_STATE CAMERA_FAULT_STATE	
Argument 1	Parameter Name:	IMAGE_INDEX
	Type:	uint16_t
	Length (bytes):	2
	Description:	Image index. Range from 0 to 767 where the first 0 to 511 indexes will point to DDR4 16GB and 512 to 767 indexes will refer to DDR4 8GB.
	Valid Values:	0x0000 to 0x02FF
Argument 2	Parameter Name:	NUM_IMAGES

	Type:	uint16_t
	Length (bytes):	2
	Description:	Number of images to transfer. If the requested number of images is larger than the mmax allowable in the memory buffer an error will be returned. If index + requested number of images is greater than max index range an error will be returned
	Valid Values:	0x0001 to 0x0300
Return Value	None	
Command Specific Errors	PLACEHOLDER	

5.3.2.12 RESET_BUFF_SYS_CMD

NAME	VALUE	
Description	Resets the buffer write pointer to a specific memory location.	
Opcode	0x0B	
Accepted Modes	CAMERA_LP_STATE CAMERA_IDLE_STATE CAMERA_ARMED_STATE	
Invalid Modes	CAMERA_BUSY_STATE CAMERA_FAULT_STATE	
Argument 1	Parameter Name:	BUFF_START_ADDR
	Type:	uint16_t
	Length (bytes):	2
	Description:	Image index. Range from 0 to 767 where the first 0 to 511 indexes will point to DDR4 16GB and 512 to 767 indexes will refer to DDR4 8GB.
	Valid Values:	0x0000 to 0x02FF
Return Value	None	
Command Specific Errors	PLACEHOLDER	

5.3.2.13 GET_INFO_SYS_CMD

NAME	VALUE	
Description	Resets the buffer write pointer to a specific memory location.	
Opcode	0x0C	
Accepted Modes	CAMERA_IDLE_STATE CAMERA_ARMED_STATE	

Invalid Modes	CAMERA_LP_STATE CAMERA_BUSY_STATE CAMERA_FAULT_STATE	
Argument 1	Parameter Name:	REG_ADDR
	Type:	uint16_t
	Length (bytes):	2
	Description:	Target Register Address
	Valid Values:	Register addresses listed in SEC REF
Return Value	None	
Command Specific Errors	PLACEHOLDER	

5.3.2.14CONF_EVENT_LOG_SYS_CMD

NAME	VALUE	
Description	Configure the event log	
Opcode	0x0D	
Accepted Modes	CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_LP_STATE	
Invalid Modes	CAMERA_BUSY_STATE CAMERA_FAULT_STATE	
Argument 1	Parameter Name:	LOG_LEVEL
	Type:	unit8_t
	Length (bytes):	1
	Description:	Enumeration of different log levels to record events: • DIAGNOSTIC = 0x00 • ACTIVITY_LO = 0x01 • ACTIVITY_HI = 0x02 • WARNING_LO = 0x03 • WARNING_HI = 0x04 • FATAL = 0x05 • COMMAND = 0x06 All event with a value equal to or higher than the provided LOG_LEVEL will be recorded to FARSIGHT's circular non-volatile TL event log for later request through XFER_LOG_SYS_CMD.
	Valid Values:	0x00 – 0x06
Return Value	None	

Command Specific Errors **PLACEHOLDER**

5.3.2.15EN_TL_CHAN_LOG_SYS_CMD

c	VALUE	
Description	Command to configure LOG buffer.	
Opcode	0x0E	
Accepted Modes	CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_LP_STATE	
Invalid Modes	CAMERA_BUSY_STATE CAMERA_FAULT_STATE	
Argument 1	Parameter Name:	CHANNEL_ADDR
	Type:	uint16_t
	Length (bytes):	2
	Description:	Target Register Address
	Valid Values:	Register addresses listed in SEC REF
Argument 2	Parameter Name:	LOG_RATE
	Type:	uint8_t
	Length (bytes):	1
	Description:	LOG_RATE in 1/10 Hz increments. This channel will be logged to FARSIGHT's circular non-volatile TL channel log for later request through XFER_LOG_SYS_CMD
	Valid Values:	0x00 – 0xFF
Return Value	None	
Command Specific Errors	PLACEHOLDER	

5.3.2.16XFER_LOG_SYS_CMD

NAME	VALUE	
Description	Request transfer of LOG file from FARSIGHT Avionics NVM to Payload computer	
Opcode	0x0F	
Accepted Modes	CAMERA_IDLE_STATE CAMERA_ARMED_STATE CAMERA_LP_STATE	
Invalid Modes	CAMERA_BUSY_STATE CAMERA_FAULT_STATE	
Argument 1	Parameter Name:	LOG
	Type:	uint8_t

	Length (bytes):	1
	Description:	Enumeration of different log levels to record events: - TL_EVENT_LOG = 0x01 (only transfers the event log) - TL_CHANNEL_LOG = 0x02 (only transfers the channel log) - FULL_LOG = 0x03 – 0xFF (transfers the entire log, maximum of 2 MB)
	Valid Values:	0x00 – 0x06
Return Value	None	
Command Specific Errors	PLACEHOLDER	

5.4 FARSIGHT REGISTERS

FARSIGHT registers are organized into several groups for convenience. These registers live in a single contiguous memory space where each address represents the previous register's address + the previous registers size. Within the tables below, the following attributes of each register are provided: address, size, data type, R/W access and if the value persists across reboots. For additional details on each register, please reference the registers individual section within section **5.6 REGISTER DESCRIPTIONS**. Data types assigned to registers are described below for convenience:

TYPE	EXAMPLES	DESCRIPTION
uint<bit_depth>_t	uint8_t, uint16_t	Unsigned integer with number of bits = bit_depth
int<bit_depth>_t	int8_t, int16_t	Signed integer with number of bits = bit_depth
uint<bit_depth>_t (bitfield)	uint8_t (bitfield)	Bitfield where each bit has a specific meaning
uint<bit_depth>_t (enum)	uint8_t (enum)	Enumeration where each value has a specific meaning
uint<bit_depth>_t[array_size]	uint8_t[4], uint32_t[2]	Unsigned integer array of size array_size with each value having number of bits = bit_depth
int<bit_depth>_t[array_size]	int8_t[4], int32_t[2]	Signed Integer Array of size array_size with each value having number of bits = bit_depth

5.4.1 CAMERA STATUS AND CONTROL REGISTERS

REGISTER NAME	ADDRESS	SIZE (BYTES)	DATA TYPE	R/W	SAVED?
SENSOR_STAT_SYS_REG	0x00	4	uint32_t	RO	No
SENSOR_FRM_MOD_SYS_REG	0x01	4	uint32_t	R/W	No
SENSOR_BIT_DEPTH_SYS_REG	0x02	4	uint32_t	R/W	No
SENSOR_GAIN_SYS_REG	0x03	4	uint32_t	R/W	No
SENSOR_BLO_SYS_REG	0x04	4	uint32_t	R/W	No
SENSOR_EXPO_USEC_SYS_REG	0x05	4	uint32_t	R/W	No

SENSOR_TEMP_KEL_SYS_REG	0x06	4	uint32_t	RO	No
SENSOR_OVER_TEMP_KEL_SYS_REG	0x07	4	uint32_t	R/W	No

5.4.2 SYSTEM REGISTERS

REGISTER NAME	ADDRESS	SIZE (BYTES)	DATA TYPE	R/W	SAVED?
ERROR_RESET_REG	0x09	4	uint32_t		
FPGA_VOLTAGE_TEMP_SYS_REG	0x0A	4	uint32_t	RO	
PWR_STATUS_REG	0x0B	4	uint32_t	RO	
ERROR_COUNT_SYS_REG	0x0C	4	uint32_t	RO	Yes
BOOT_COUNT_SYS_REG	0x0D	4	uint32_t	RO	Yes
UPTIME_SEC_SYS_REG	0x03	4	uint32_t	RO	
TIMESTAMP_UNIX_SYS_REG	0x0F	8	uint32_t[2]	RO	No
WDG_TIMEOUT_MSEC_SYS_REG	0x10	4	uint32_t	R/W	Yes
BUFF_WRITE_POLICY_SYS_REG	0x10	4	uint32_t	R/W	No
BUFF_WRITE_PTR_SYS_REG	0x11	4	uint32_t	RO	No
BUFF_BYTES_AVAIL_L_SYS_REG	0x12	4	uint32_t		
BUFF_BYTES_AVAIL_M_SYS_REG	0x13	4	uint32_t		

5.4.3 SYSTEM REGISTERS

REGISTER NAME	ADDRESS	SIZE (BYTES)	DATA TYPE	R/W	SAVED?
TRIG_MODE_SYS_REG	0x14	4	uint32_t	R/W	No
IMG_PER_TRIGGER_SYS_REG	0x15	4	uint32_t	R/W	No
FRAMES_PER_SEC_SYS_REG	0x16	4	uint32_t	R/W	No
CAP_START_DLY_USEC_SYS_REG	0x17	4	uint32_t	R/W	No
FRAME_COUNT_SYS_REG	0x18	4	uint32_t	RO	Yes
FOCUS_REQ_DIST_M_SYS_REG	0x19	4	uint32_t	RO	No

FOCUS_COMP_DIST_M_SYS_REG	0x1A	4	uint32_t	RO	No
FOCUS_LVDT_POS_NM_REG	0x1B	4	int32_t	RO	No
IMAGE_BUFFER_SELECT_REG	0x1C	4	uint32_t		

5.4.4 SYSTEM COMMUNICATION REGISTERS

REGISTER NAME	ADDRESS	SIZE (BYTES)	DATA TYPE	R/W	SAVED?
GPO_CONF_SYS_REG	0x1D	4	uint32_t	R/W	No
GPI_CONF_SYS_REG	0x1E	4	uint32_t	R/W	No
INTR_PIN_CONF_SYS_REG	0x1F	4	uint32_t	R/W	No
TRIG_PIN_CONF_SYS_REG	0x20	4	uint32_t	R/W	No
TRIG_DELAY_USEC_SYS_REG	0x21	4	uint32_t	R/W	No
TRIG_TIMEOUT_MSEC_SYS_REG	0x22	4	uint32_t	R/W	No
TMTC_TIMEOUT_MSEC_SYS_REG	0x23	4	uint32_t	R/W	No
SRC_MAC_ADDRESS_0_3_SYS_REG	0x24	4	uint32_t		
SRC_MAC_ADDRESS_4_5_SYS_REG	0x25	4	uint32_t		
DST_MAC_ADDRESS_0_3_SYS_REG	0x26	4	uint32_t		
DST_MAC_ADDRESS_4_5_SYS_REG	0x27	4	uint32_t		
SRC_IP_ADDRESS_SYS_REG	0x28	4	uint32_t		
DEST_IP_ADDRESS_SYS_REG	0x29	4	uint32_t		
GATEWAY_SYS_REG	0x2A	4	uint32_t	R/W	No
SUBNET_MASK_SYS_REG	0x2B	4	uint32_t	R/W	No
SRC_PORT_SYS_REG	0x2C	4	uint32_t		
DST_PORT_SYS_REG	0x2D	4	uint32_t		

5.4.5 SYSTEM VERSION REGISTERS

REGISTER NAME	ADDRESS	SIZE (BYTES)	DATA TYPE	R/W	SAVED?
---------------	---------	--------------	-----------	-----	--------

FAV_DEVICE_ID_SYS_REG	0x2E	4	uint32_t	RO	Yes
SENSOR_ID_SYS_REG	0x2F	4	uint32_t	RO	Yes
TMTC_VERS_SYS_REG	0x30	4	uint32_t	RO	Yes
SW_VERS_SYS_REG	0x31	4	uint32_t	RO	Yes
PF_FW_VERS_SYS_REG	0x32	4	uint32_t	RO	Yes
P3_FW_VERS_SYS_REG	0x33	4	uint32_t	RO	Yes

5.4.6 TEST & INTEGRATION

REGISTER NAME	ADDRESS	SIZE (BYTES)	DATA TYPE	R/W	SAVED?
ENABLE_IMX_TEST_IMG	0x34	4	uint32_t		
NOOP_REG	0x35	4	uint32_t	R/W	No
NUMBER_REGISTERS	0x36	4	uint32_t	RO	Yes

5.5 CAMERA STATUS & CONTROL REGISTER DESCRIPTIONS

This section includes tabulated descriptions of FARSIGHT user registers. Reference [sections 5.4.1 - 5.4.6](#) for address, R/W access and if the register's value persists across resets.

5.5.1 SENSOR_STAT_SYS_REG

Description	The current status of FARSIGHT's camera sensor.		
Access	Read Only		
Address	0x00		
Data Type	unit32_t		
Reset Value	0x0000_0000 (DISABLED)		
Enumeration Values	Value	Enumeration	Description
	0x0000_0000	DISABLED	When the sensor is currently disabled and cannot initial an acquisition
	0x0000_0001	STANDBY	When the sensor can be configured but is not ready to take images
	0x0000_0002	ARMED	Sensor is capable of capturing images
	0x0000_0003	N/A	This bit is not applicable and will result in returned error

5.5.2 SENSOR_FRM_MOD_SYS_REG

Description	Configuration defining frame modification included in default sensor features			
Access	Read/Write			
Address	0x01			
Data Type	unit32_t (bitfield)			
Reset Value	0x00 (see below)			
Bitfield Description	Bitfield Value	Index	Description	
	RESERVED	31:4	Reserved bits	
	READOUT_MODE	3:2	2-bit enumerated value, values described below:	
			Value	Meaning
			0	All-pixel Scan
			1	2x2 binning
			2	½ subsampling
			3	All-pixel scan
	VERT_INVERSION	1	If set to 1, then the image will have its vertical axis inverted	
	HORIZ_INVERSION	0	If set to 1, then the image will have its horizontal axis inverted	

5.5.3 SENSOR_BIT_DEPTH_SYS_REG

Description	The bit-depth of the images that FARSIGHT's sensor will output. <i>This parameter has been identified as a future improvement for FARSIGHT. Until further notice, this bit-depth will be locked at 12-bits.</i>
Access	Read/Write
Address	0x02

Data Type	uint32_t		
Reset Value	0x0000_0002 (TWELVE_BITS)		
Enumeration Values	Value	Enumeration	Description
	0x0000_0000	EIGHT_BITS	Each pixel is 8-bits wide
	0x0000_0001	TEN_BITS	Each pixel is 10-bits wide
	0x0000_0002	TWELVE_BITS	Each pixel is 12-bits wide

5.5.4 SENSOR_GAIN_SYS_REG

Description	Gain in dB represented by an integer 10x the desired gain. (see Format)		
Access	Read/Write		
Address	0x03		
Data Type	unit32_t		
Reset Value	0		
Minimum	0		
Maximum	480		
Format	When the user sets this value, the sensor will divide the value by 10. Example: User sets SENSOR_GAIN_SYS_REG to 345, resulting gain is 34.5 dB.		

5.5.5 SENSOR_BLO_SYS_REG

Description	Black level offset configuration		
Access	Read/Write		
Address	0x04		
Data Type	unit32_t		
Reset Value	240 (This value is recommended for 12-bit operation)		
Minimum	0		
Maximum	4095		

5.5.6 SENSOR_EXPO_USEC_SYS_REG

Description	Exposure in microseconds (TBR)		
Access	Read/Write		
Address	0x05		
Data Type	uint32_t		
Reset Value	500 (TBR)		
Minimum	0 – represents minimum exposure time of the sensor (50 nanoseconds) (TBR)		
Maximum	2 ²²		

5.5.7 SENSOR_TEMP_KEL_SYS_REG

Description	Raw 16-bit ADC reading of the image sensor's thermometer in Kelvin (TBR)		
Access	Read Only		

Address	0x06
Data Type	uint32_t
Reset Value	0x00
Minimum	0x00
Maximum	65535 (TBR)

5.5.8 SENSOR_OVER_TEMP_KEL_SYS_REG

Description	Raw 16-bit ADC reading marking an over-temperature event. If this condition persists for 1 second (TBR), then a transition to LP_MODE will automatically be triggered. Setting this value to 65535 will disable this feature.
Access	Read/Write
Address	0x07
Data Type	uint32_t
Reset Value	0x00
Minimum	0x00
Maximum	65535

5.5.9 ERROR_RESET_REG

Description	
Access	
Address	0x08
Data Type	uint32_t
Reset Value	
Minimum	
Maximum	

5.5.10 FPGA_VOLTAGE_TEMP_SYS_REG

Description	
Access	
Address	0x09
Data Type	
Reset Value	
Minimum	
Maximum	

5.5.11 PWR_STATUS_REG

Description	
Access	Read Only
Address	0x0A

Data Type	uint32_t
Reset Value	
Minimum	
Maximum	

5.5.12 ERROR_COUNT_SYS_REG

Description	Total number of errors encountered by FARSIGHT
Access	Read Only
Address	0x0B
Data Type	uint32_t
Reset Value	0
Minimum	0
Maximum	4294967295

5.5.13 BOOT_COUNT_SYS_REG

Description	Total number of times FARSIGHT was reset or power cycled
Access	Read Only
Address	0x0C
Data Type	uint32_t
Reset Value	0
Minimum	0
Maximum	4294967295

5.5.14 UPTIME_MSEC_SYS_REG

Description	
Access	
Address	0x0D
Data Type	uint32_t
Reset Value	
Minimum	
Maximum	

5.5.15 TIMESTAMP_UNIX_SYS_REG

Description	FARSIGHT's timestamp using the Unix epoch (January 1, 1970)	
Access	Read Only (Writeable through SET_TIME_SYS_CMD)	
Address	0x0E	
Data Type	uint32[2]	
Field Descriptions	Index	Description
	0	Seconds since Epoch

	1	Sub-Seconds (nanoseconds)
Reset Value	0,0	
Minimum	0,0	
Maximum	0xFFFFFFFF,0xFFFFFFFF	

5.5.16 WDOG_TIMEOUT_MSEC_SYS_REG

Description	Internal software watchdog timeout in milliseconds for FARSIGHT's. If this timer expires, then FARSIGHT will revert to CAMERA_LP_STATE. A value of 4294967295 disables this timeout feature.		
Access	Read/Write		
Address	0x0F		
Data Type	uint32_t		
Reset Value	4294967295		
Minimum	0		
Maximum	4294967295		

5.5.17 BUFF_WRITE_POLICY_SYS_REG

Description	Dictates how FARSIGHT will handle buffer overflow in the event that the frame buffer fills up during imaging.		
Access	Read/Write		
Address	0x10		
Data Type	uint32_t		
Reset Value	CIRCULAR (0x00)		
Enumeration Values	Value	Enumeration	Description
	0x0000_0000	CIRCULAR	Frame buffer is circular, overwriting older images with newer images
	0x0000_0001	OVERWRITE_LAST	Overwrite the last image collected
	0x0000_0002	STOP_IMAGING	Stop imaging if the buffer fills up

5.5.18 BUFF_WRITE_PTR_SYS_REG

Description	Current address of the write pointer for that FARSIGHT's FPGA is using to write to the DDR4 frame buffer		
Access	Read Only		
Address	0x11		
Data Type	uint32_t		
Reset Value	0		
Minimum	0		
Maximum			

5.5.19 BUFF_BYTES_AVAIL_L_SYS_REG

Description	
Access	
Address	0x12
Data Type	uint32_t
Reset Value	
Minimum	
Maximum	

5.5.20 BUFF_BYTES_AVAIL_M_SYS_REG

Description	
Access	
Address	0x13
Data Type	uint32_t
Reset Value	
Minimum	
Maximum	

5.5.21 TRIG_MODE_SYS_REG

Description	Configuration of FARSIGHT's image trigger mode. Based on this value, FARSIGHT will watch for the respective trigger event in CAMERA_ARMED_STATE prior to initiating image capture.		
Access	Read/Write		
Address	0x14		
Data Type	uint32_t		
Reset Value	COMMAND (0x00)		
Enumeration Values	Value	Enumeration	Description
	0x0000_0000	COMMAND	Start exposure immediately after receiving a START_ACQ_SYS_CMD command.
	0x0000_0001	EXT_GPIO	Start exposure immediately after a configured GPIO's rising edge has been detected.
	0x0000_0002	PARAM_UPDATE	Consider an update to SENSOR_GAIN_SYS_REG, SENSOR_BLO_SYS_REG and/or SENSOR_EXPO_USEC_SYS_REG as a request to image.
	0x0000_0003	TIME	Start exposure after transition to either MANUAL_MODE or STREAM_MODE once a configured amount of time has passed. To configure the amount of time to wait, set the CAP_START_DLY_USEC_SYS_REG.

5.5.22 IMG_PER_TRIGGER_SYS_REG

Description	When in CAMERA_BUSY_STATE FARSIGHT will take the number of images set within this register after every trigger event has been detected (see section 5.6.1.21 TRIG_MODE_SYS_REG).
Access	Read/Write
Address	0x15
Data Type	uint32_t
Reset Value	1
Minimum	1
Maximum	500 (TBR)

5.5.23 FRAMES_PER_SEC_SYS_REG

Description: Frame rate when capturing image sequences, only used when in fast imaging mode

Description	Frame rate of image capture in frames/second after a trigger was detected in CAMERA_ARMED_STATE. Higher frame rates are achievable when in ½ sampling or 2x2 binning modes (see section 5.6.1.2 SENSOR_FRM_MOD_SYS_REG). <i>Note: methods to calculate subsampling frame rates will be included in future releases of FARSIGHT's Software ICD.</i>			
Access	Read/Write			
Address	0x16			
Data Type	uint32_t			
Reset Value	1			
Minimum	1			
Maximum	Bit Depth	8-bit	10-bit	12-bit
	Max Frame Rate	109	104	75

5.5.24 CAP_START_DLY_USEC_SYS_REG

Description	Value in microseconds used to configure when FARSIGHT starts taking images after successful transition to CAMERA_BUSY_STATE and when TRIG_MODE_SYS_REG is set to TIME (see section 5.6.1.21 TRIG_MODE_SYS_REG).
Access	Read/Write
Address	0x17
Data Type	uint32_t
Reset Value	0
Minimum	0
Maximum	4294967295 (TBR)

5.5.25 FRAME_COUNT_SYS_REG

Description	Rolling counter to associate with successfully captured frames. After each frame this is incremented and included in the metadata for an image.
-------------	---

Access	Read Only
Address	0x18
Data Type	uint32_t
Reset Value	0
Minimum	0
Maximum	4294967295

5.5.26 FOCUS_REQ_DIST_M_SYS_REG

Description	Focus distance requested by the user from a FOCUS_SYS_CMD, which is the only way to update this register value.
Access	Read Only
Address	0x19
Data Type	uint32_t
Reset Value	N/A
Minimum	5000
Maximum	10000000

5.5.27 FOCUS_COMP_DIST_M_SYS_REG

Description	Focus distance computed from LVDT position and calibrated LUT
Access	Read Only
Address	0x1A
Data Type	uint32_t
Reset Value	Computed on startup after the first LVDT read
Minimum	5000
Maximum	10000000

5.5.28 FOCUS_LVDT_POS_NM_REG

Description	Reading of the sensor position in nanometers that can be mapped to focus distance.
Access	Read Only
Address	0x1B
Data Type	uint32_t
Reset Value	Read on startup
Minimum	TBD
Maximum	TBD

5.5.29 BUFF_BYTES_AVAIL_M_SYS_REG

Description	
Access	0x1C
Address	uint32_t

Data Type	
Reset Value	
Minimum	
Maximum	

5.5.30 GPO_CONF_SYS_REG

Description	Configuration for programmable general purpose output pin. <i>Note: details towards configuring actions when FARSIGHT detects this pin asserted will be included in future releases of FARSIGHT's software ICD.</i>	
Access	Read/Write	
Address	0x1D	
Data Type	uint32_t	
Reset Value	0x00	
Minimum	0x00	
Bitfield Description	Bitfield Value	Index
	RESERVED	31:4
	Source enum (TBD)	3:2
	Active Low (if not set, assumed active high)	1
	Enable Pin	0

5.5.31 GPI_CONF_SYS_REG

Description	Configuration for programmable general purpose input pin. <i>Note: details towards configuring events that lead to FARSIGHT asserting this pin will be included in future releases of FARSIGHT's software ICD.</i>	
Access	Read/Write	
Address	0x1E	
Data Type	uint32_t	
Reset Value	0x00	
Minimum	0x00	
Bitfield Description	Bitfield Value	Index
	RESERVED	31:4
	Event enum (TBD)	3:2
	Active Low (if not set, assumed active high)	1
	Enable Pin (enable or disable this feature – 1 = enable)	0

5.5.32 INTR_PIN_CONF_SYS_REG

Description	Configuration for the programmable interrupt pin that can be asserted for various events such as FRAME_READY, TRANSFER_COMPLETE, etc. <i>Note: details towards configuring events that lead to FARSIGHT asserting this pin will be included in future releases of FARSIGHT's software ICD.</i>
-------------	--

Access	Read/Write	
Address	0x1F	
Data Type	uint32_t	
Reset Value	0x00	
Minimum	0x00	
Bitfield Description	Bitfield Value	Index
	RESERVED	31:4
	Event enum 0x00 = FRAME_COMPLETE 0x01 = MODE_TRANSITION 0x02 = TBD 0x03 = TBD	3:2
	Active Low (if not set, assumed active high)	1
	Enable Pin (enable or disable this feature – 1 = enable)	0

5.5.33 TRIG_PIN_CONF_SYS_REG

Description	Configuration for FARSIGHT's image trigger pin that will lead to triggering of image captures when FARSIGHT's TRIG_MODE_SYS_REG is set to HOST_PIN.	
Access	Read/Write	
Address	0x20	
Data Type	uint32_t	
Reset Value	0x00	
Minimum	0x00	
Bitfield Description	Bitfield Value	Index
	RESERVED (TBR)	31:1
	Active Low (if not set, assumed active high)	0

5.5.34 TRIG_DELAY_USEC_SYS_REG

Description	Number of microseconds for FARSIGHT to wait before initiating image capture after a trigger event has been detected. A value of 0 effectively disables this feature.	
Access	Read/Write	
Address	0x21	
Data Type	uint32_t	
Reset Value	0	
Minimum	0	
Maximum	4294967295	

5.5.35 TRIG_TIMEOUT_MSEC_SYS_REG

Description	Internal timeout in milliseconds (TBR) for FARSIGHT to wait for a trigger event after transitioning to MODE_MANUAL or MODE_STREAM. If this timeout occurs, then FARSIGHT will transition back to IDLE_MODE. (see section 5.6.1.21 TRIG_MODE_SYS_REG). A value of 4294967295 disables this timeout feature.
Access	Read/Write
Address	0x22
Data Type	uint32_t
Reset Value	4294967295
Minimum	0
Maximum	4294967295

5.5.36 TMTC_TIMEOUT_MSEC_SYS_REG

Description	Internal timeout in milliseconds for FARSIGHT's packet deserializer to wait for a complete packet after receiving a valid header or sync frame. A value of 4294967295 disables this timeout feature.
Access	Read/Write
Address	0x23
Data Type	uint32_t
Reset Value	4294967295
Minimum	0
Maximum	4294967295

5.5.37 SRC_MAC_ADDRESS_0_3_SYS_REG

Description	
Access	
Address	0x24
Data Type	uint32_t
Reset Value	
Minimum	
Maximum	

5.5.38 SRC_MAC_ADDRESS_4_5_SYS_REG

Description	
Access	
Address	0x25
Data Type	uint32_t
Reset Value	
Minimum	

Maximum

5.5.39 DST_MAC_ADDRESS_0_3_SYS_REG

Description	
Access	
Address	0x26
Data Type	uint32_t
Reset Value	
Minimum	
Maximum	

5.5.40 DST_MAC_ADDRESS_4_5_SYS_REG

Description	
Access	
Address	0x27
Data Type	uint32_t
Reset Value	
Minimum	
Maximum	

5.5.41 SRC_IP_ADDRESS_SYS_REG

Description	
Access	
Address	0x28
Data Type	uint32_t
Reset Value	
Minimum	
Maximum	

5.5.42 DEST_IP_ADDRESS_SYS_REG

Description	
Access	
Address	0x29
Data Type	uint32_t
Reset Value	
Minimum	
Maximum	

5.5.43 GATEWAY_SYS_REG

Description	FARSIGHT's IPv4 gateway address
Access	Read/Write
Address	0x2A
Data Type	uint32_t[4]
Reset Value	

5.5.44 SUBNET_MASK_SYS_REG

Description	FARSIGHT's IPv4 address
Access	Read/Write
Address	0x2B
Data Type	uint32_t[4]
Reset Value	

5.5.45 SRC_PORT_SYS_REG

Description	
Access	
Address	0x2C
Data Type	uint32_t
Reset Value	
Minimum	
Maximum	

5.5.46 DST_PORT_SYS_REG

Description	
Access	
Address	0x2D
Data Type	uint32_t
Reset Value	
Minimum	
Maximum	

5.5.47 FAV_DEVICE_ID_SYS_REG

Description	Unique hardware revision number for FARSIGHT's Avionics board
Access	Read Only
Address	0x2E
Data Type	uint32_t

5.5.48 SENSOR_ID_SYS_REG

Description	Unique hardware revision number for FARSIGHT's Sensor board
Access	Read Only
Address	0x2F
Data Type	uint32_t

5.5.49 TMTC_VERS_SYS_REG

Description	TMTC Protocol revision number	
Access	Read Only	
Address	0x30	
Data Type	uint32_t[2]	
Field Descriptions	Index	Description
	0	Major revision number
	1	Minor revision number

5.5.50 SW_VERS_SYS_REG

Description	Revision number for software running on FARSIGHT's RISC-V softcore	
Access	Read Only	
Address	0x31	
Data Type	uint32_t[2] (TBR)	
Field Descriptions	Index	Description
	0	Major revision number
	1	Minor revision number

5.5.51 PF_FW_VERS_SYS_REG

Description	Revision number for firmware flashed on FARSIGHT's PolarFire FPGA	
Access	Read Only	
Address	0x32	
Data Type	uint32_t[2] (TBR)	
Field Descriptions	Index	Description
	0	Major revision number
	1	Minor revision number

5.5.52 P3_FW_VERS_SYS_REG

Description	Revision number for firmware flashed on FARSIGHT's ProASIC3 FPGA	
Access	Read Only	
Address	0x33	
Data Type	uint32_t (TBR)	

5.5.53 ENABLE_IMX_TEST_IMG

Description	
Access	
Address	0x34
Data Type	uint32_t (TBR)
Reset Value	
Minimum	
Maximum	

5.5.54 NOOP_REG

Description	
Access	Read/Write
Address	0x34
Data Type	uint32_t (TBR)
Reset Value	
Minimum	
Maximum	

5.5.55 NUMBER_REGISTERS

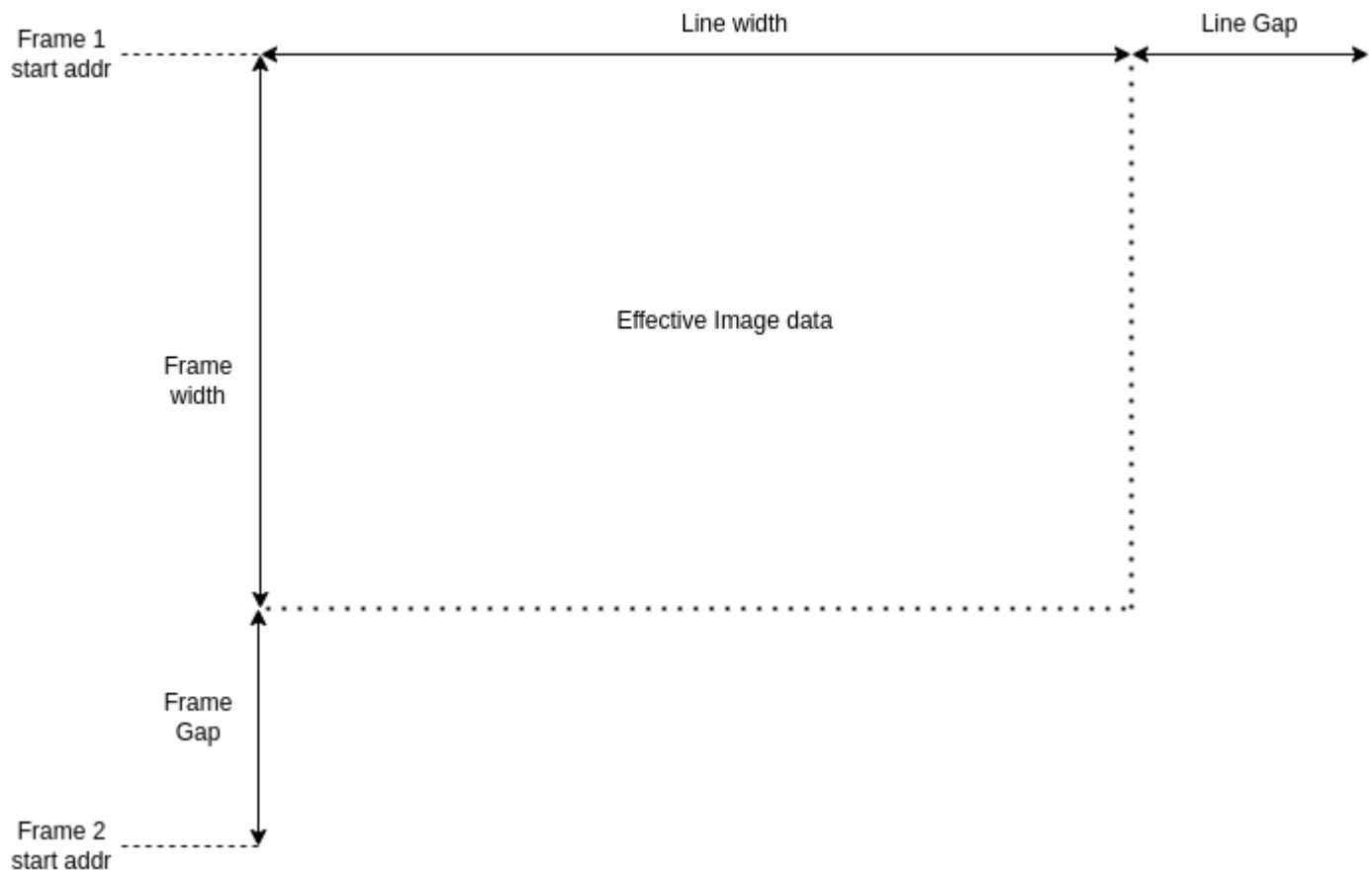
Description	
Access	Read/Only
Address	0x34
Data Type	uint32_t (TBR)
Reset Value	
Minimum	
Maximum	

5.0 PCIe DATA TRANSFER

Details on PCIe data transfer will be included in future releases of FARSIGHT's software ICD.

6.0 IMAGE FORMAT & METADATA

6.1 IMAGE FRAME FORMAT



The total DDR4 memory is 24 GB, addressed using a 35-bit counter. This counter is split: the upper 9 bits select the frame number, and the lower 26 bits select the location within the frame. Each frame starts with a metadata line, followed by pixel data lines. Line addresses increase by a configurable step size (e.g., 12 KB). The lines are incremented by a given constant value. The Frame index is incremented on each frame valid signal assertion.

While using UDP data is provided in Row format without extra Line Gaps in the data

If PCIe is used with the start address and transfer size the transferred frame will include line gaps which needs to be processed on the receive side.

6.2 IMAGE METADATA

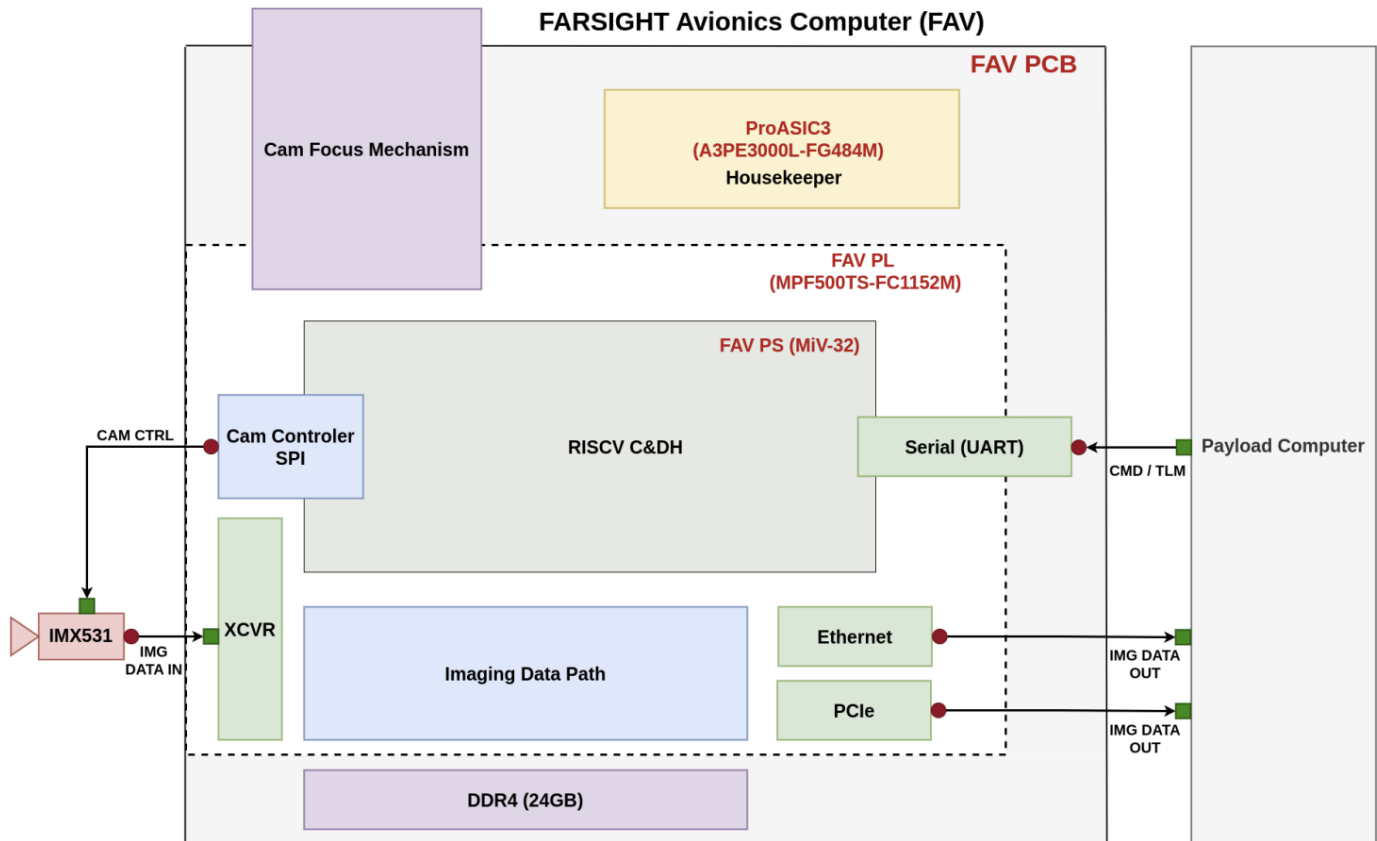
FARSIGHT's metadata header is detailed in the table below:

Field	Size (bytes)	Type	Byte Address	Default Value(s)	Description
METADATA_START_FLAG	4		0x00	0x4D45_5441	Start flag
METADATA_SIZE_BYTES			0x04		Total metadata block size in bytes
SENSOR_PXL_READOUT_FORMAT	4	uint32_t	0x08		Sensor frame output format
SENSOR_READ_DIR	4	uint32_t	0x0C		Sensor readout direction

SENSOR_BIT_DEPTH	4	uint32_t	0x10		Sensor image bit-depth
SENSOR_GAIN	4	uint32_t	0x14		Sensor gain dB (X10)
SENSOR_BLO	4	uint32_t	0x18		Sensor black level offset
SENSOR_EXPO_USEC	4	uint32_t	0x1C		Exposure time in microseconds
SENSOR_TRIG_MODE	4	uint32_t	0x20		Image trigger mode configuration
SENSOR_TEMP_RAW	4	uint32_t	0x24		Raw sensor temperature reading
BUFF_WRITE_INDEX	4	uint32_t	0x28		DDR4 frame buffer write pointer index
FOCUS_REQ_DIST_M	4	uint32_t	0x2C		Requested focus distance
FOCUS_COMP_DIST_M	4	uint32_t	0x30		Computed focus distance from LVDT
FOCUS_LVDT_POS_NM	4	uint32_t	0x34		LVDT sensor position in nanometer
TIMESTAMP_UNIX_EPOCH_SEC	4	uint32_t	0x38		Unix epoch timestamp in seconds
TIMESTAMP_SUBSEC_NSEC	4	uint32_t	0x3C		Sub-seconds timestamp in nanoseconds
VERSION	4	uint32_t (bitfield)	0x40		Subsystem version info
RESERVED PADDING			0x44		Reserved zero padding
RESERVED PADDING			0x48		Reserved zero padding
METADATA-CRC32			0x4C		CRC32 of Metadata calculated by FPGA during capture

Appendix A: FARSIGHT High-level Hardware Design

Fig Ref shows the high level architecture of FARSIGHT Payload system.



The FARSIGHT Avionics Computer (FAV) is the interface between the Payload Computer and the Camera Sensor (IMX531). The FARSIGHT electronics reside on the FAV PCB which consists of two main FPGA chips and supporting components:

- **ProASIC3 (A3PE3000L-FG484M) – Housekeeper FPGA:** The housekeeper FPGA monitors power fluctuations and performs latchup protection in case of single-event latchup (SEL) issues caused by radiation
- **PolarFIRE (MPF500TS-FC1152M) – FAV PL:** The PolarFire FPGA serves as the main control interface and contains two primary subsystems
 - **FAV PS (MiV-32):** The Processing System hosts the RISC-V MiV-32 softcore processor which acts as the main Command and Data Handling (C&DH) unit. It is responsible for receiving and processing telecommands from the Payload Computer and responding with telemetry information.
 - **Imaging Data Path:** The programmable logic section handles high-speed image data processing and frame buffer management.

The main Telemetry and Telecommand (TMTC) interface between Farsight Avionics and the Payload Computer is a Serial UART interface backed by RS422 differential signaling for reliable CMD/TLM communication.

The Camera Sensor interface to the system consists of two main paths: the control path and the data path.

- **Control Path (CAM CTRL):** The Cam Controller uses a SPI interface that allows configuration of the IMX531 camera sensor via SPI protocol.
- **Data Path (IMG DATA IN):** The data path interface consists of 8-lane high-speed SERDES lanes running at 4752 Mbps/lane connected to the transceivers (XCVR) of the PolarFire FPGA. The image data passes through the Imaging Data Path pipeline and is stored in frame buffers allocated in the external 24GB DDR4 memory.

Additionally, the FAV includes a Camera Focus Mechanism for controlling the sensor position relative to the optical train.

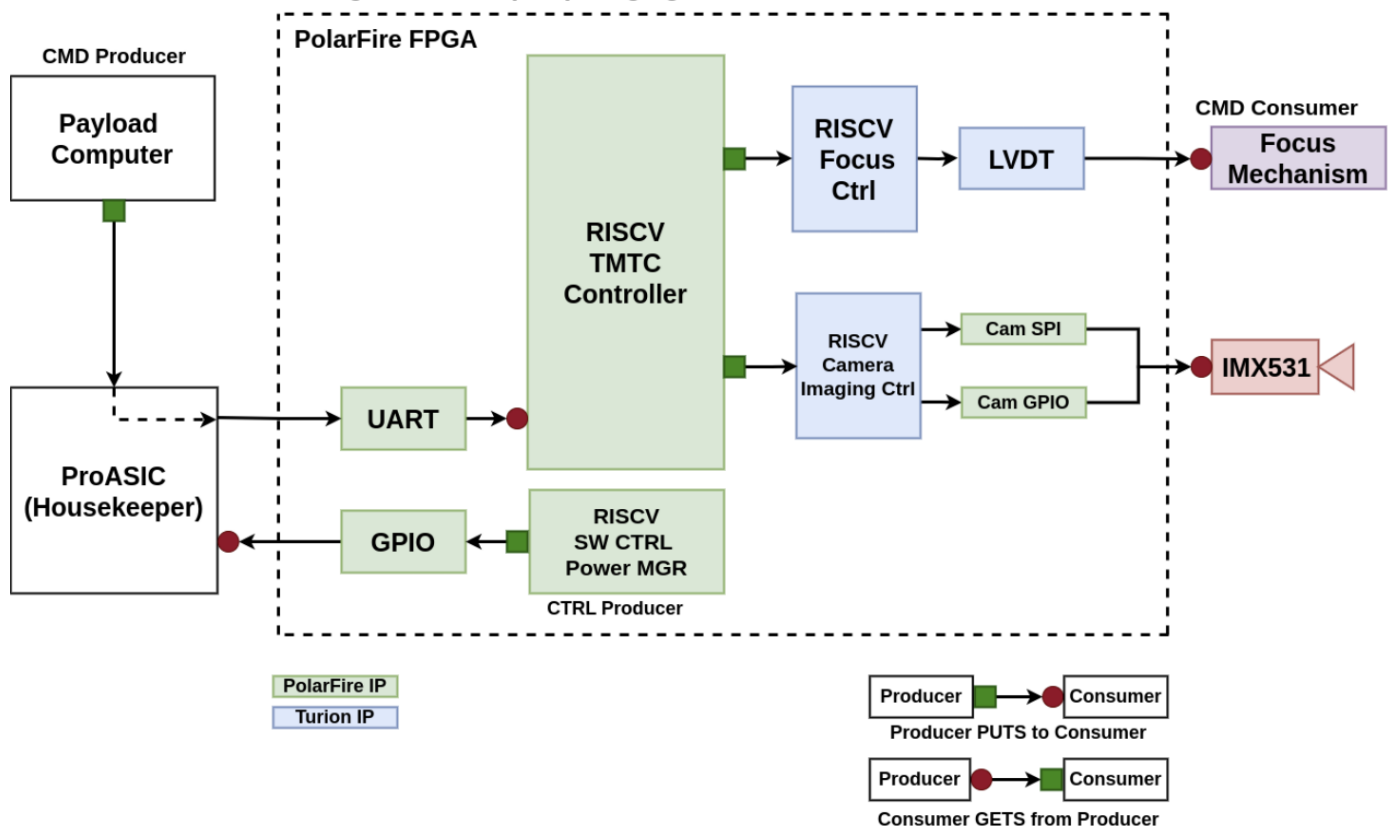
For image data output (IMG DATA OUT), there are two main options to transfer data to the Payload Computer:

- **Ethernet:** The 1 Gigabit Ethernet interface supports up to 3 fps using User Datagram Protocol (UDP).
- **PCIe:** The PCIe endpoint interface support 5 Gbps data rate using PCIe Gen2.

Image Control Path:

Fig Ref shows the high-level view of the FARSIGHT Avionics imaging control path.

Farsight Avionics (FAV) Imaging Control Path



The Payload Computer sends telecommands to the FAV through a communication path that passes through the Housekeeper FPGA. Under nominal conditions, the Housekeeper FPGA operates as a pass-through for this communication link as long as the PolarFire FPGA is powered on. In the event that the PolarFire FPGA fails to

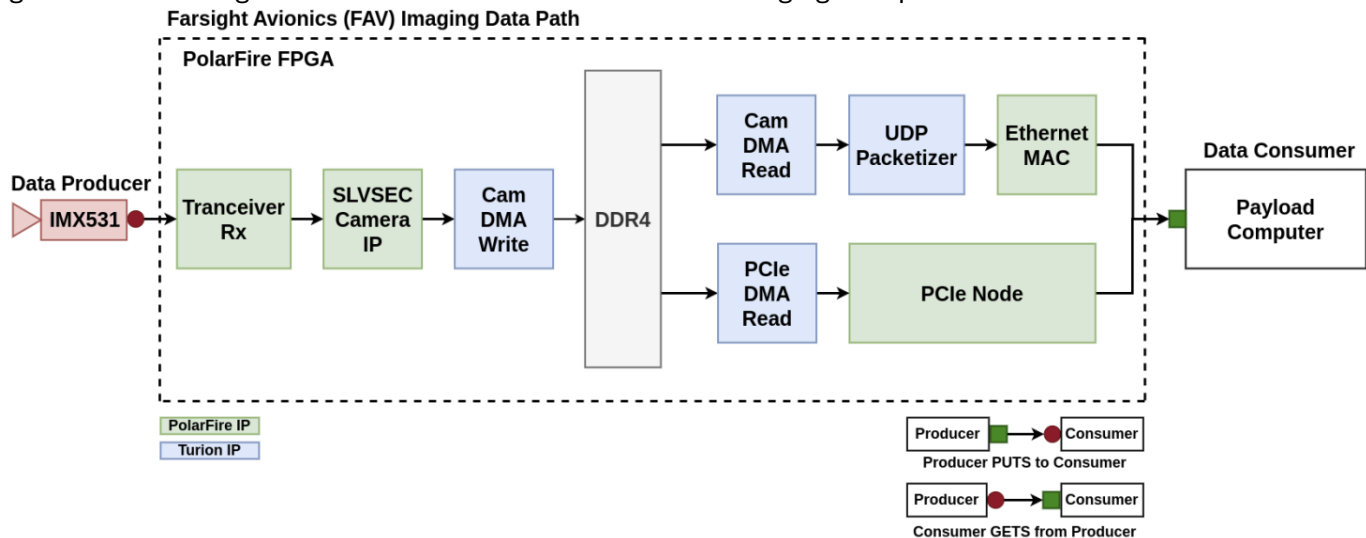
power on, the Housekeeper FPGA assumes control of this path and transmits a beacon signal back to the Payload Computer indicating which power region has failed.

During nominal operation, telecommands from the Payload Computer are received by the RISC-V processor in the PolarFire FPGA via the serial port interface. The RISC-V firmware parses these commands and dispatches them to the appropriate target subsystem. The imaging control path consists of the following controllable subsystems:

- **Camera Control Module:** The camera control module consists of two main components
 - **Camera SPI Interface:** Provides configuration access to the IMX531 camera sensor registers via SPI protocol.
 - **Trigger Module:** Generates trigger pulses to initiate image capture based on user-specified exposure settings and frame rate (frames per second) parameters.
- **Focus Mechanism Controller:** Enables precise positioning of the camera focus mechanism through telecommands to achieve optimal image focus
- **Housekeeper Interface Module:** Provides the RISC-V firmware with the ability to monitor system status and issue commands to the Housekeeper FPGA for controlling various power regions within the system.

Imaging Data Path:

Fig Ref shows the high-level view of the FARSIGHT Avionics imaging data path.



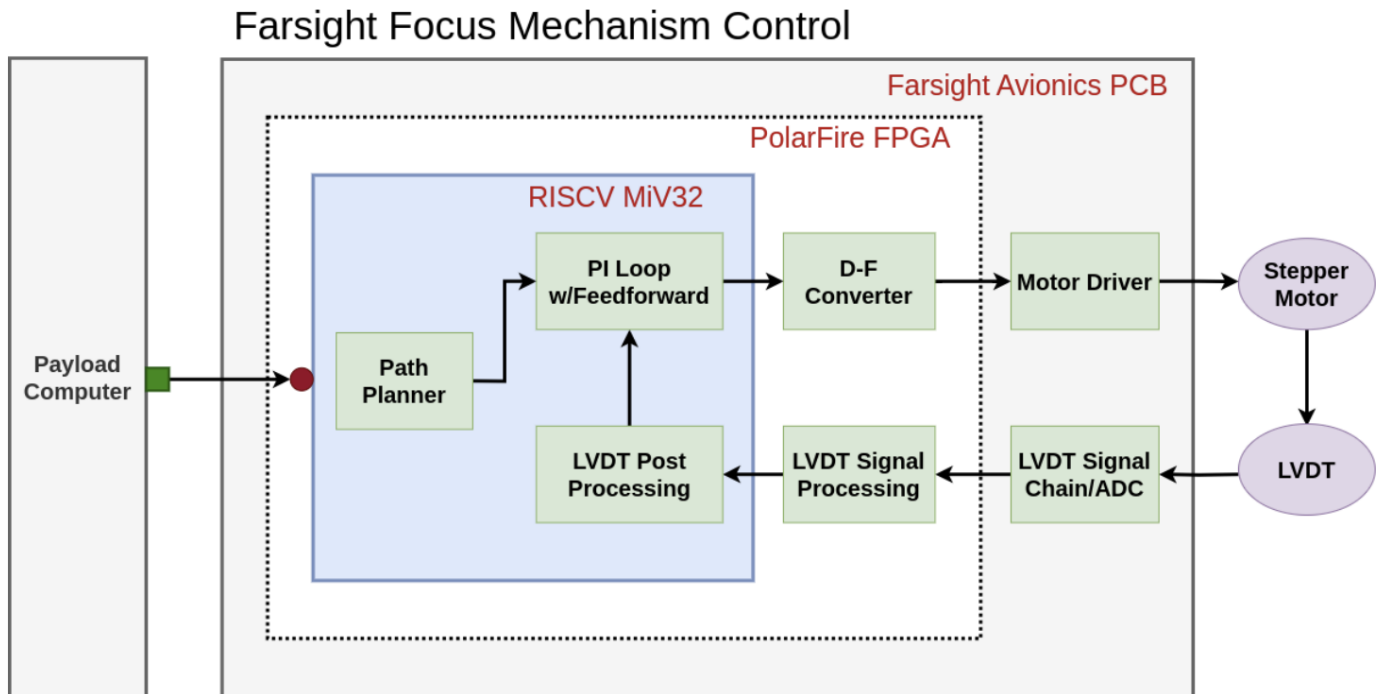
The imaging data path illustrates how image data flows from the IMX531 sensor to the Payload Computer. The IMX531 sensor acts as the data producer, transmitting image data through the Transceiver Rx block into the SLVSEC Camera IP within the PolarFire FPGA. The Camera DMA write module then stores the incoming image frames into DDR4 memory.

From DDR4 memory, the system supports two parallel data paths to the Payload Computer (data consumer):

- **Ethernet Path:** The CAM DMA Read module retrieves image data from DDR4, which is then packetized by the UDP Packetizer and transmitted through the ethernet MAC to the Payload Computer.
- **PCIe Path:** The PCIe DMA Read module retrieves image data from the DDR4 and transfers it through the PCIe Node directly to the Payload Computer

Focus Mechanism:

Fig Ref shows the focus mechanism control system architecture.



The focus mechanism control loop operates as follows: Upon receiving a focus command from the Payload Computer, the RISC-V MiV32 processor initiates path planning. If the commanded setpoint is too close to the current position, the Path Planner will first move the sensor away from the target before approaching the setpoint to avoid stiction issues.

The Path Planner generates updated velocity and position commands which are fed to the PI Loop with Feedforward. The PI loop employs velocity feedforward to improve motion profile tracking accuracy. Position feedback is provided by a Linear Variable Differential Transformer (LVDT) sensor, whose signal passes through the LVDT Signal Chain/ADC for analog-to-digital conversion, followed by LVDT Signal Processing and LVDT Post Processing blocks before being fed back to the PI loop.

The PI loop output is converted by the D-F Converter and sent to the Motor Driver, which commands the Stepper Motor to adjust the focus mechanism position. The motor is stopped once the desired position is reached, or if the position error exceeds a specified threshold in microns.

Hardware Power Protection:

FARSIGHT Avionics Housekeeper consists of two FPGA chips for managing and monitoring power status of the system. The ProASIC3 FPGA and PolarFire FPGA. The ProASIC3 FPGA Housekeeper is responsible for controlling and monitoring power to various subsystems on the Farsight Avionics board. It manages 33 power

sources organized into 11 power regions, each with dedicated state machines for sequencing and fault detection. During normal operation, the ProASIC3 operates in pass-through mode, routing UART traffic directly between the PolarFire and the payload bus. If a critical failure occurs and the PolarFire is powered down, the ProASIC3 takes control of the UART and broadcasts a failure status message approximately once per second. The message format is shown in Table Ref.

FSW Nice To Have Requests

This section is dedicated to a list of items that FSW are requesting as nice to haves

- A general SOH telemetry message. Right now, to get the important health telemetry of farsight, the FSW has to query many different registers. It would simplify operations if we could have a command to the farsight that did that for us
- It would be nice to not have to set the camera parameters every time before taking an image
- The UDP packets that are received should probably have a checksum
- It would be nice to be able to set the sensor so that it produces test images
- It would be nice if the DDR4 RAM was set to some known pattern at bootup
- It might be nice if there was some way to “clear” DDR4 memory

FSW Bug Fix and Improvement Requests

This section is dedicated to a list of items that FSW are requesting as important to have fixed changed or added

- The request image command should not have anything hardcoded and should